

Modular Polynomials Associated with Modular Parametrizations of Elliptic Curves over $\mathbb{Q}$

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Abstract

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N . We construct a class of modular polynomials $F_N(x, j)$ that relate the modular function x on $X_0(N)$ to the j -invariant j , where x is obtained by composing the first coordinate function of E with the modular parametrization $\varphi : X_0(N) \rightarrow E$. Using $F_N(x, j)$, we can precisely determine the poles of φ , compute exact values of φ at cusps, and develop an algorithm for calculating branch values of φ . Moreover, $F_N(x, j)$ yields an effective algorithm for computing the fibers of φ over arbitrary points on E . $F_N(x, j)$ also provides a resultant-based method for obtaining candidate minimal polynomials of the images of Heegner points on $X_0(N)$ under φ . In particular, we compute the semi-trace of the image $\varphi([\frac{-1+\sqrt{-3}}{2}])$ of the CM-point $[\frac{-1+\sqrt{-3}}{2}]$ on $X_0(389)$, under the action of a 65-element subgroup of the 260-element Galois group of $\mathbb{Q}(\sqrt{-3}, j(389 \cdot \frac{-1+\sqrt{-3}}{2}))$. As a further application, we take the reverse viewpoint and show that a rational point of infinite order on E gives rise, via the fibers of φ , to an infinite sequence $\{(j(\tau_n), j(N\tau_n))\}_{n \geq 1}$ of algebraic points on $Z_0(N)$ whose degrees are bounded by $\deg \varphi$. This construction provides an algebraic way to encode rational points on E in terms of modular data.

Keywords. Modular parametrization; elliptic curve; modular curve; modular polynomial; CM point; computational number theory

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1 Introduction

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N , and fix a modular parametrization

$$\varphi : X_0(N) \longrightarrow E, \quad \varphi(P_\infty) = O_E.$$

Its existence follows from the modularity theorem [21, 20, 3]. Writing $x = x_E \circ \varphi$ and $y = y_E \circ \varphi$, we study how to determine algebraic relations for these coordinate functions and use them to compute fibres and distinguished images on E . The functions x and y may have poles away from the cusps, so their algebraic relations retain information that is not visible from the cusp images alone.

Put $J(\tau) = j(N\tau)$. We construct primitive irreducible integer-coefficient relations

$$F_N(x, j) = 0, \quad f_N(x, J) = 0, \quad G_N(y, j) = 0, \quad g_N(y, J) = 0.$$

The construction uses exact q -expansions at ∞ , degree bounds, and rational linear algebra. A finite coefficient test based on pole-divisor degrees proves the resulting identities, and the same test selects the irreducible factor that actually vanishes on the modular curve. The construction does not rely on selecting a factor solely by its degree or coefficient height. For other constructions of modular and class polynomials, see [4, 5, 12].

Griffin and Hales [13] study divisors and fibres of modular parametrizations using products over $\mathrm{SL}_2(\mathbb{Z})$ -conjugates of rational functions in x and y . Their recursive expansions at the cusps permit computations of selected symmetric coefficient functions, expressed as rational functions of j , and they also study CM criteria and congruences. Their displayed examples provide selected coefficients and information about particular fibres, rather than a complete bivariate coordinate polynomial $F_N(x, j)$. Here the output consists of the complete coordinate relations above, constructed from expansions at a

single cusp and accompanied by a finite identity certificate. This is a distinction in the algebraic output and its verification, not a claim of lower computational cost.

The principal new outputs are an exact example of non-CM zeros of an Eichler integral for 38a1, and an explicit semi-trace in a quadratic field for 389a1. The first follows from a certified coordinate polynomial and the period map; the second has a compact principal-divisor certificate for the specified sum of conjugate points. The coordinate polynomials also provide algebraic candidates for fibres, cusp images, branch values, and coordinates of images of CM points. We distinguish candidate generation from the exact recognition of points on the smooth modular curve, including points above singularities of a plane model.

The computation of critical and ramification points goes back to Mazur and Swinnerton-Dyer [18]. Delaunay [11] developed a numerical method for locating zeros of the associated newform, and Chen [7] introduced the Poly Relation and Poly Relation-YP algorithms for critical polynomials. Brunault [6] and Corbett and Saha [9] determine ramification at cusps. Our branch-value procedure is an application of the coordinate polynomials already constructed: resultants supply a finite search set, and complete fibres decide which candidates are branch values. It includes separate treatment of cusps, the exceptional modular invariants 0, 1728, and the fibre over O_E . We make no performance comparison with these earlier algorithms.

Kodgis [16, Section 5, p. 10] asked whether all zeros in $\mathcal{H}$ of the relevant Eichler integrals occur at CM points. Peluse [19, Theorem 1.1] proved sufficient conditions for several families of CM zeros. She also reports Ono's speculation that CM zeros describe the generic behaviour [19, p. 72]. Our 38a1 example gives a negative answer to Kodgis's universal question; it does not settle a statement about generic behaviour.

We use $\mathcal{H}^* = \mathcal{H} \cup \mathbb{Q} \cup \{\infty\}$, $X_0(N) = \Gamma_0(N) \backslash \mathcal{H}^*$, and $Y_0(N) = \Gamma_0(N) \backslash \mathcal{H}$. The class of τ is denoted by $[\tau]$. We write

$$F_N(x, j) = \sum_{k=0}^K \sum_{l=0}^L c_{k,l} x^k j^l = \sum_{k=0}^K A_k(j) x^k = \sum_{l=0}^L B_l(x) j^l. \quad (1)$$

Here $K = \deg_x F_N$, $L = \deg_j F_N$, $\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$, and $d = \deg \varphi$. The notation F_N also depends on the chosen curve and parametrization, which remain fixed in each example. By a pole of a coordinate function we mean a point of $\varphi^{-1}(O_E)$; the morphism φ itself is defined everywhere on $X_0(N)$.

A CM point represents a cyclic isogeny between elliptic curves whose endomorphism rings are orders in the same imaginary quadratic field; we call it a Heegner point when the two orders agree [14, 24]. If Q is such a point, choose a finite Galois extension $L/\mathbb{Q}$ containing its residue field. Then $\varphi(Q) \in E(L)$, and, for a subgroup $H \leq \mathrm{Gal}(L/\mathbb{Q})$, its semi-trace is

$$S_H = \sum_{\sigma \in H} \sigma(\varphi(Q)) \in E(L^H).$$

Section 7 proves the requisite field of definition for its specific CM point. A class polynomial combined with a coordinate polynomial provides coordinate candidates by elimination; selecting the correct factor and embedding is a separate step.

Section 2 gives the construction, certificate, and experimental information. Sections 3–6 treat fibres, the fibre over O_E , cusp images, and branch values. Section 7 contains the semi-trace example, and Section 8 explains degree drops and coordinate symmetries. Section 9 recovers the coordinates as rational functions in generators of the modular function field. Section 10 records the direct bounded-degree consequence of taking fibres over rational points.

2 Algorithm for Constructing Modular Polynomials $F_N(x, j)$

2.1 Modular parametrizations and coordinate functions

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N , given by a Weierstrass equation

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6. \quad (2)$$

Let

$$\omega_E = \frac{dx}{2y + a_1 x + a_3}$$

be the associated invariant differential. Let

$$\varphi : X_0(N) \longrightarrow E$$

be a modular parametrization of E , normalized by $\varphi(P_\infty) = O_E$. Let f be the normalized newform of weight 2 on $\Gamma_0(N)$ associated with E .

For $\tau \in \mathcal{H}$, define the Eichler integral

$$\gamma(\tau) = \int_{i\infty}^{\tau} 2\pi i f(z) dz. \quad (3)$$

At cusps, the value of γ is understood as the corresponding limiting value whenever the limit exists.

Throughout the paper, unless otherwise stated, we assume for simplicity that E is the optimal curve in its isogeny class and that its Manin constant is 1. Under this assumption,

$$\varphi^* \omega_E = 2\pi i f(z) dz. \quad (4)$$

Let Λ_E be the period lattice of E with respect to ω_E . Then the Abel–Jacobi coordinate of $\varphi([\tau])$ is represented by $\gamma(\tau)$ modulo Λ_E . Moreover, for any fixed $\tau \in \mathcal{H}$, the quantities

$$\gamma(M\tau) - \gamma(\tau), \quad M \in \Gamma_0(N),$$

are periods of E , and the period lattice is generated by these modular periods:

$$\Lambda_E = \langle \gamma(M\tau) - \gamma(\tau) : M \in \Gamma_0(N) \rangle_{\mathbb{Z}}. \quad (5)$$

The lattice obtained in this way is independent of the choice of τ . The optimal quotient $J_0(N) \rightarrow E$ has connected kernel and therefore induces a surjection on integral first homology. Using the Abel–Jacobi identification of $H_1(X_0(N), \mathbb{Z})$ with $H_1(J_0(N), \mathbb{Z})$, and the Manin-constant-one normalization, we obtain the stated equality of period lattices.

Let $b_2 = a_1^2 + 4a_2$, and let $\wp = \wp_{\Lambda_E}$ be the Weierstrass $\wp$ -function attached to the lattice Λ_E . Under the Manin-constant one normalization, the coordinates of $\varphi([\tau])$ are given by

$$x_\varphi(\tau) = \wp(\gamma(\tau)) - \frac{b_2}{12}, \quad (6)$$

$$y_\varphi(\tau) = \frac{\wp'(\gamma(\tau))}{2} - \frac{a_1 \wp(\gamma(\tau))}{2} + \frac{a_1 b_2 - 12a_3}{24}. \quad (7)$$

Equivalently,

$$\varphi([\tau]) = (x_\varphi(\tau), y_\varphi(\tau)).$$

The functions x_φ and y_φ descend to meromorphic functions on $X_0(N)$. Thus they may be regarded as maps

$$x_\varphi, y_\varphi : X_0(N) \longrightarrow \mathbb{P}^1(\mathbb{C}).$$

In particular, if x_φ is finite at a point of $X_0(N)$, then x_φ is holomorphic there; the same statement holds for y_φ .

Since $f = q + O(q^2)$, the chosen normalization gives

$$\gamma = q + O(q^2), \quad x = q^{-2} + O(q^{-1}), \quad y = -q^{-3} + O(q^{-2}).$$

The minus sign follows from $\wp'(u) = -2u^{-3} + O(u)$. The q -expansion of f can be computed effectively, and hence the Eichler integral $\gamma(\tau)$ can be evaluated term by term for $\tau \in \mathcal{H}$. The formulas above therefore give an effective analytic procedure for evaluating $\varphi([\tau])$. For further details on these computations, see [8, 11].

We now define the polynomial $F_N(X, Y)$. Let

$$J(\tau) = j(N\tau).$$

It is classical that the function field of $X_0(N)$ over $\mathbb{C}$ is

$$\mathbb{C}(X_0(N)) = \mathbb{C}(j, J).$$

Since x_φ is a meromorphic function on $X_0(N)$, we have

$$x_\varphi \in \mathbb{C}(X_0(N)) = \mathbb{C}(j, J).$$

In particular, x_φ is algebraic over $\mathbb{C}(j)$. Hence there exists a nonzero polynomial

$$F_N(X, Y) \in \mathbb{C}[X, Y]$$

such that

$$F_N(x_\varphi, j) = 0.$$

Because the modular parametrization φ is defined over $\mathbb{Q}$, the function x_φ has a q -expansion at the cusp ∞ with rational coefficients. It follows that the algebraic relation between x_φ and j may be chosen over $\mathbb{Q}$. After clearing denominators and dividing by the greatest common divisor of the coefficients, we obtain a primitive polynomial

$$F_N(X, Y) \in \mathbb{Z}[X, Y],$$

unique up to sign after choosing the irreducible factor determined by x_φ and j . We call $F_N(X, Y)$ the x -coordinate modular polynomial associated with the modular parametrization φ . For brevity, we also refer to it as the x -coordinate polynomial associated with φ . In the notation of the paper, we write this relation as

$$F_N(x, j) = 0,$$

where x denotes the modular function $x_\varphi = x_E \circ \varphi$.

Let

$$\mu = [SL_2(\mathbb{Z}) : \Gamma_0(N)],$$

and let

$$\{M_n : 1 \leq n \leq \mu\}$$

be a complete set of right coset representatives for $\Gamma_0(N) \backslash SL_2(\mathbb{Z})$. Finally, denote by

$$d = \deg(\varphi)$$

the degree of the modular parametrization.

Proposition 2.1. $K \leq \mu$ and $L \leq 2d$.

Proof. Since $j : X_0(N) \rightarrow X(1) \simeq \mathbb{P}^1$ has degree $\mu = [SL_2(\mathbb{Z}) : \Gamma_0(N)]$, we have

$$[\mathbb{C}(X_0(N)) : \mathbb{C}(j)] = \mu.$$

The field $\mathbb{C}(x, j)$ is a subfield of $\mathbb{C}(X_0(N))$. Therefore

$$K = \deg_x F_N(x, j) = [\mathbb{C}(x, j) : \mathbb{C}(j)] \leq \mu.$$

On the other hand, the map

$$x = x_E \circ \varphi : X_0(N) \longrightarrow \mathbb{P}^1$$

has degree $2 \deg(\varphi)$, since the x -coordinate map $E \rightarrow \mathbb{P}^1$ has degree 2. Hence

$$L = \deg_j F_N(x, j) = [\mathbb{C}(x, j) : \mathbb{C}(x)] \leq 2 \deg(\varphi) = 2d.$$

□

2.2 Construction by q -expansions and linear algebra

Let

$$x(\tau) = \sum_{n=-2}^{\infty} a_n q^n, \quad j(\tau) = \sum_{n=-1}^{\infty} b_n q^n, \quad x^k j^l = \sum_{n=-2k-l}^{\infty} c(k, l; n) q^n,$$

where $q = e^{2\pi i \tau}$. Then

$$\sum_{k=0}^K \sum_{l=0}^L c_{k,l} \sum_{n=-2k-l}^{\infty} c(k, l; n) q^n \equiv 0.$$

Thus

$$\sum_{n=-2K-L}^{\infty} \left(\sum_{k=0}^K \sum_{l=0}^L c_{k,l} c(k, l; n) \right) q^n \equiv 0.$$

This implies for all $n \geq -2K - L$,

$$\sum_{k=0}^K \sum_{l=0}^L c_{k,l} c(k, l; n) = 0.$$

Thus the coefficients $c_{k,l}$ can be determined by solving this system of linear equations. Then we have the following algorithm.

Algorithm 1 Construction of Modular Polynomial $F_N(x, j)$

Input: Elliptic curve E over $\mathbb{Q}$ with conductor N , with the modular parametrization φ fixed as above.
Output: The primitive irreducible polynomial $F_N(X, Y) \in \mathbb{Z}[X, Y]$ satisfying $F_N(x, j) = 0$, with positive leading coefficient in the fixed lexicographic order.

- 1: Compute μ and $d = \deg(\varphi)$; set $K := \mu$ and $L := 2d$. Set $R := (K + 1)(L + 1)$ and $M := R + 20$.
- 2: Compute exact q -expansions of x and j to a precision sufficient to determine all products $x^k j^l$, $0 \leq k \leq K$, $0 \leq l \leq L$, through the coefficient of q^M .
- 3: For each pair (k, l) , compute

$$x^k j^l = \sum_{n=-2K-L}^M c(k, l; n) q^n + O(q^{M+1}).$$

- 4: Solve the linear system over $\mathbb{Q}$

$$\sum_{k=0}^K \sum_{l=0}^L c_{k,l} c(k, l; n) = 0, \quad -2K - L \leq n \leq M.$$

Choose a nonzero solution and form the candidate $H(X, Y) := \sum_{k=0}^K \sum_{l=0}^L c_{k,l} X^k Y^l$.

- 5: Set $B := 2dK + \mu L$. Extend the input expansions as needed and check exactly that

$$[q^n]H(x(q), j(q)) = 0, \quad -2K - L \leq n \leq B.$$

If this check fails, set $M := \max\{2M, B\}$ and repeat from Step 2. If it succeeds, Lemma 2.5 certifies $H(x, j) \equiv 0$.

- 6: Factor H over $\mathbb{Q}$ as

$$H = c \prod_{i=1}^r H_i^{e_i}, \quad c \in \mathbb{Q}^\times,$$

where the $H_i \in \mathbb{Z}[X, Y]$ are distinct primitive irreducible factors, up to sign. Set $k_i := \deg_X H_i$, $l_i := \deg_Y H_i$, and $B_i := 2dk_i + \mu l_i$.

- 7: Select the unique factor H_i for which

$$[q^n]H_i(x(q), j(q)) = 0, \quad -2k_i - l_i \leq n \leq B_i.$$

Return $F_N := H_i$, with positive leading coefficient in the fixed lexicographic order.

Remark 2.2. (i) In Step 1, the degree d of φ can be computed using the `ellmoddegree` function in PARI/GP; (ii) In Step 2, the q -expansion of x can be computed using the `elltaniyama` function in PARI/GP; (iii) With minor modifications to Algorithm 1, one can compute the modular polynomials $f_N(x, J)$, $G_N(y, j)$, and $g_N(y, J)$.

Example 2.3. The computational results for $N = 11$ yield

$$\begin{aligned} F_{11}(x, j) = & (16 - x)^{11} j^2 \\ & + (-104748564078368391 + 199736619430410535x + 159480622275659333x^2 \\ & + 6839041777752481x^3 - 29669709666741936x^4 - 4074814667347831x^5 \\ & + 1134855511654843x^6 + 164063633585170x^7 + 5072626276355x^8 \\ & + 38323813979x^9 + 43119747x^{10} + 1486x^{11})j \\ & + (9789217 + 4971236x + 1333262x^2 - 52820x^3 + x^4)^3, \end{aligned}$$

$$f_{11}(x, J) = (16 - x)J^2 + (6969 + 5732x - 12529x^2 - 6105x^3 + 1309x^4 + 297x^5 - 22x^6)J \\ + (97 + 116x + 62x^2 - 20x^3 + x^4)^3.$$

Remark 2.4. *The assumptions that E is optimal and that the Manin constant is 1 are made only to simplify the analytic normalization. If the Manin constant of φ is c_E , then $\varphi^*\omega_E = c_E 2\pi i f(z) dz$, and the Abel–Jacobi coordinate is represented by $c_E \gamma(\tau)$ rather than by $\gamma(\tau)$. If E is not optimal, one may first work with the optimal curve E_0 in the same isogeny class and then compose its modular parametrization with the rational isogeny $E_0 \rightarrow E$. In either case, the coordinate functions obtained from φ remain meromorphic modular functions on $X_0(N)$. Thus these assumptions are not essential for the construction of the polynomial $F_N(X, Y)$.*

2.3 Finite q -expansion certificate

The linear algebra construction described above produces a candidate polynomial relation. The truncation used to obtain such a candidate is only a device for finding a likely relation. To prove that the candidate is an actual identity in the function field of $X_0(N)$, we use the standard divisor-degree criterion for meromorphic functions on a compact Riemann surface.

Lemma 2.5 (Finite q -expansion certificate for F_N). *Let $X = X_0(N)$, and let P_∞ be the cusp ∞ . Let*

$$H(X, Y) = \sum_{k=0}^K \sum_{l=0}^L c_{k,l} X^k Y^l$$

be a candidate polynomial relation, and put

$$h = H(x, j) \in \mathbb{C}(X).$$

Then

$$(h)_\infty \leq K(x)_\infty + L(j)_\infty.$$

In particular,

$$\deg(h)_\infty \leq K \deg(x) + L \deg(j).$$

Consequently, if

$$\text{ord}_{P_\infty} h > K \deg(x) + L \deg(j),$$

then

$$H(x, j) \equiv 0$$

as a meromorphic function on $X_0(N)$.

In the normalization used above,

$$x(q) = q^{-2} + O(q^{-1}), \quad j(q) = q^{-1} + 744 + O(q).$$

Thus $H(x(q), j(q))$ has no terms below degree

$$-2K - L.$$

Therefore it is enough to verify

$$[q^n] H(x(q), j(q)) = 0 \quad \text{for all } -2K - L \leq n \leq B_F,$$

where

$$B_F = K \deg(x) + L \deg(j).$$

Since $x = x_E \circ \varphi$, where $x_E : E \rightarrow \mathbb{P}^1$ has degree 2, we have

$$\deg(x) = 2d,$$

with $d = \deg \varphi$. Moreover

$$\deg(j) = \mu = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)].$$

Hence one may take

$$B_F = 2dK + \mu L.$$

Proof. For each monomial $x^k j^l$, its pole divisor is bounded by

$$(x^k j^l)_\infty \leq k(x)_\infty + l(j)_\infty \leq K(x)_\infty + L(j)_\infty.$$

Since the pole divisor of a sum is bounded by the maximum of the pole divisors of its terms, we obtain

$$(h)_\infty \leq K(x)_\infty + L(j)_\infty.$$

Hence

$$\deg(h)_\infty \leq K \deg(x) + L \deg(j).$$

If $h \neq 0$, then the zero and pole divisors of h have the same degree on the compact Riemann surface $X_0(N)$:

$$\deg(h)_0 = \deg(h)_\infty.$$

But if

$$\text{ord}_{P_\infty} h > K \deg(x) + L \deg(j),$$

then the zero at P_∞ alone contributes more than the total possible pole degree, a contradiction. Hence $h \equiv 0$. $\square$

The same argument gives the corresponding finite certificates for the other polynomial relations. For $f_N(x, J)$, where

$$J(\tau) = j(N\tau),$$

one has

$$\deg(J) = \mu,$$

and hence the global bound is

$$B_f = K \deg(x) + L \deg(J) = 2dK + \mu L.$$

Since

$$J(q) = q^{-N} + 744 + O(q^N),$$

the local starting exponent is

$$-2K - NL.$$

Thus it is enough to check

$$[q^n] H(x(q), J(q)) = 0 \quad \text{for all } -2K - NL \leq n \leq 2dK + \mu L.$$

For the next two certificates, K, L denote the bidegree bounds of the respective candidate in its first and second variables. For $G_N(y, j)$, since

$$y(q) = -q^{-3} + O(q^{-2}), \quad \deg(y) = 3d,$$

the local starting exponent is

$$-3K - L,$$

and the global bound is

$$B_G = 3dK + \mu L.$$

Thus it is enough to check

$$[q^n] H(y(q), j(q)) = 0 \quad \text{for all } -3K - L \leq n \leq 3dK + \mu L.$$

For $g_N(y, J)$, the local starting exponent is

$$-3K - NL,$$

and the global bound is

$$B_g = 3dK + \mu L.$$

Thus it is enough to check

$$[q^n] H(y(q), J(q)) = 0 \quad \text{for all } -3K - NL \leq n \leq 3dK + \mu L.$$

In Algorithm 1, M is the highest exponent used for construction, whereas $R = (K + 1)(L + 1)$ is the number of formal unknowns. We start with $M = R + 20$, giving

$$M + 2K + L + 1 = R + 2K + L + 21$$

coefficient equations. The additional terms are a search heuristic; only the stated finite certificate proves an identity. For example, exact unit series q^2x and qj , each known modulo $q^{M+2K+L+1}$, suffice to determine every product $x^k j^l$ through q^M . The input precision for the final certificate is increased in the same way with M replaced by its required highest exponent.

Finally, the certificate proves that the candidate is a genuine relation, but does not by itself identify the irreducible relation F_N . Given a certified nonzero polynomial $H \in \mathbb{Q}[X, Y]$, factor it over $\mathbb{Q}$ as

$$H = c \prod_{i=1}^r H_i^{e_i}, \quad c \in \mathbb{Q}^\times,$$

where the $H_i \in \mathbb{Z}[X, Y]$ are distinct primitive irreducible factors, up to sign. For each factor, put

$$k_i = \deg_X H_i, \quad l_i = \deg_Y H_i, \quad B_i = 2dk_i + \mu l_i,$$

and apply Lemma 2.5 to check exactly that

$$[q^n]H_i(x(q), j(q)) = 0 \quad \text{for all } -2k_i - l_i \leq n \leq B_i.$$

The unique factor passing this check is F_N , up to sign. Indeed, the substitution homomorphism

$$\mathbb{Q}[X, Y] \longrightarrow \mathbb{Q}(X_0(N)), \quad P \longmapsto P(x, j),$$

has a height-one prime kernel generated by F_N . Since H belongs to this kernel, exactly one of its distinct irreducible factors belongs to the kernel. Coefficient height and bidegree are used only to select candidates; the factor is identified by the certified vanishing condition.

The degree bounds $K = \mu$ and $L = 2d$ ensure that the linear system in Algorithm 1 always has a nonzero solution. If the construction precision is increased until $M \geq B = 2dK + \mu L$, every solution already satisfies the full certificate. Thus the precision increase terminates in exact arithmetic. Here M and B are highest exponents: checking through q^B means establishing $H(x(q), j(q)) = O(q^{B+1})$.

Certification and output conventions. Polynomial coefficients and algebraic input points are represented exactly; an algebraic number is accompanied by an isolating region specifying its complex embedding. All polynomial relations used in later certificates are required to pass the finite tests above. A rational reconstruction from decimal values is only a candidate until it passes an exact check or is uniquely isolated in a proved finite candidate set. The fibre procedures in Sections 3–6 use certified enclosures with propagated error bounds, as specified in Section 3, and return representatives of distinct points on $X_0(N)$, not merely distinct pairs of plane coordinates. For unrestricted complex input, these procedures describe the mathematical fibres without asserting an exact terminating implementation for an arbitrary complex-number representation. The verification status of the displayed computations is recorded separately in Section 2.4 and the supplement.

2.4 Experimental questions and reproducible checks

The experiments address three questions: can a complete coordinate relation be constructed and certified; what arithmetic information can be certified from it; and which additional checks are needed when plane coordinates do not distinguish points of the smooth modular curve? The examples are selected for these mathematical roles. Table 1 states the selection and the scope of the available evidence. This is a validation collection drawn from the examples in the article, not a statistical sample or a census of curves in a conductor interval.

Representation size. For a primitive integer polynomial $F = \sum c_{k,l} X^k Y^l$, we record the number $s(F)$ of nonzero coefficients and the maximum decimal digit count

$$h_{10}(F) = \max_{c_{k,l} \neq 0} (\lfloor \log_{10} |c_{k,l}| \rfloor + 1).$$

Curve	Experimental question	Evidence and scope
11a1	Can the construction be reproduced from the curve equation?	From-scratch construction and finite certification of F_{11} .
38a1	Can a relation prove an arithmetic statement about the pole fibre?	Complete F_{38} certificate and the non-CM-zero proof in Section 4.
40a1	How does a symmetry change the relation degrees?	The reported degrees give $r = 2$; numerical fibre labels require the recognition of Section 3.
46a1	How are coincident plane coordinates and branch candidates distinguished?	Section 6 separates the candidate conditions from the pending complete fibre certificates. The Yang-pair identity in Section 9 has its own exact certificate.
91b1	Are the reported fibre invariants integral?	The displayed polynomial is primitive, irreducible, and nonmonic; the polynomial check does not certify its fibre assignment.
176a1	Can cusps contribute ramification in the fibre over O_E ?	Section 5 records the cusp fibre and indices using the cited cusp criterion; no new numerical-image certificate is claimed here.
389a1	Can a large sum of CM conjugates be independently checked?	A compact certificate for the specified semi-trace, conditional on the supplied F_{389} being the coordinate relation.

Table 1: Selection of examples and the conclusions supported by their evidence.

Relation	μ	d	K	L	r	s	h_{10}	Bytes
F_{11}	12	1	12	2	1	37	22	841
F_{38}	60	6	60	12	1	706	125	65449

Table 2: Two complete certified relations under the same coefficient encoding.

Signs are excluded from this digit count. To make byte counts unambiguous, we use an uncompressed UTF-8 JSON array of triples $[\mathbf{k}, \mathbf{l}, \mathbf{c}]$, sorted by increasing (k, l) , containing only nonzero coefficients. The coefficient is its signed decimal string; there are no spaces or trailing newline. Content is removed and the last coefficient in this ordering is made positive.

Here $r = [\mathbb{C}(X_0(N)) : \mathbb{C}(x, j)]$, so $K = \mu/r$ and $L = 2d/r$, as explained in Section 8. Recording r is essential: when $r > 1$, the relation curve is not a birational model of $X_0(N)$. Table 2 describes the two outputs; it gives neither a growth law nor a comparison with classical modular polynomials under the same encoding. The sparsity of a final polynomial also does not show that its support was known before construction.

What the large rectangular model counts. For the 389a1 parameters reported in the article,

$$K = 390, \quad L = 80, \quad d = 40, \quad \mu = 390,$$

the rectangular ansatz has

$$R = (K + 1)(L + 1) = 31671$$

formal unknown coefficients. The sufficient certificate bound is

$$B_F = 2dK + \mu L = 62400,$$

and the range from $-2K - L = -860$ through B_F , inclusively, contains 63261 coefficient positions. With the initial choice $M = R + 20 = 31691$ in Algorithm 1, the construction system has $M + 2K + L + 1 = 32552$ rows before dependencies are removed. These are counts for the stated rectangular model, not measurements of matrix rank, stored entries, running time, or memory.

The reported relation degrees give $r = 1$ in this example. They do not determine whether an implementation exploited additional support constraints or used modular arithmetic or parallel blocks. Those choices must be identified from the construction code and logs. In particular, the certificate need not store a dense matrix with 63261 rows: it can evaluate the given polynomial and check the specified coefficients. Its resource use nevertheless requires a measurement, and the full F_{389} certificate has not been rerun in the validation collection below.

Measured construction and verification tasks. The supplementary driver fixes five tasks before execution and runs them sequentially in fresh PARI/GP 2.15.2 processes. The available machine reports an AMD EPYC 9V74 processor and Linux x86-64. Each GP process uses `nthreads=1` and an initial PARI stack of 256,000,000 bytes. One run per task was recorded; cache state and other host activity were not controlled.

Elapsed time is measured by a monotonic wall clock from process launch through termination and includes loading, parsing, checking, and output. The polling interval is approximately 0.01 seconds, so elapsed times below 0.1 seconds are reported only as such. Peak RSS is the maximum resident set size of that child process, obtained from Linux `wait4`; 1 MiB means 2^{20} bytes. It is distinct from the initial PARI stack setting and from memory allocated to a job. The JSON record additionally contains user and system CPU times, executable hash, and environment information.

Task	Wall time (s)	Peak RSS (MiB)	Status
Construct and certify F_{11} from the curve	< 0.1	21.1	PASS
Load and certify the complete F_{38} relation	11.67	197.7	PASS
Load and check the 389a1 semi-trace certificate	0.35	21.1	PASS
Construct and check the 46a1 Yang-pair certificate	< 0.1	21.1	PASS
Check the 91b1 polynomial	< 0.1	21.1	PASS

Table 3: Single-run resource measurements for the fixed validation tasks.

For 11a1, we construct the relation directly from exact q -expansions of x and j , using the model $y^2 + y = x^3 - x^2 - 10x - 20$, without loading a precomputed relation. Here $K = 12$, $L = 2$, and $R = 39$. We take $M = B_F = 48$, where M denotes the highest exponent checked in the substituted relation, in place of the heuristic initial value $M = R + 20 = 59$ in Algorithm 1. The input expansions are computed to sufficient precision to account for the negative powers in the products. Checking the coefficients from q^{-26} through q^{48} gives a matrix with 75 rows and 39 columns, whose kernel over $\mathbb{Q}$ is one-dimensional. The finite q -expansion bound certifies its generator as a relation. If this relation were reducible over $\mathbb{Q}$, one of its proper factors would also vanish at (x, j) , yielding a second linearly independent kernel vector within the same bidegree bounds. Thus, after primitive normalization, the generator is the minimal relation, and no separate bivariate factorization is required.

The remaining tasks verify specified outputs or identities. In particular, the 389a1 entry in Table 3 measures verification of the semi-trace certificate. Construction and full verification of F_{389} are outside the scope of that measurement. These runs document individual computational tasks; they do not establish how construction costs grow with the conductor or quantify the amortized cost of repeated fibre or CM queries.

The original construction of F_{389} required approximately 6.1 TB of peak memory and 120 hours of wall-clock time. These figures concern a separate construction run and should be distinguished from the semi-trace verification measurements in Table 3.

The archived notebook `F389xj.nb` occupies 21,456,072 bytes. The previously reported size of approximately 900 MB for the classical modular polynomial is a historical figure. Since the two files have not been compared using the common encoding specified in Table 2, their sizes provide only an indication of storage requirements in the respective formats. Notebook overhead, compression, and polynomial representation affect this comparison, and the size ratio does not imply a corresponding ratio of computational running times.

3 Algorithm for computing the fibre of φ over a point of E

Let $[\tau] \in X_0(N)$, $\alpha = x(\tau)$ and $\beta = j(\tau)$ such that $\alpha, \beta \in \mathbb{C}$. By the definition of $F_N(X, Y)$, we have $F_N(\alpha, \beta) = 0$. Moreover, we have the following proposition.

Proposition 3.1. *Let (α, β) be a finite affine point on the irreducible curve defined by $F_N(X, Y) = 0$. Then there exists a point $Q \in X_0(N)$ such that*

$$x(Q) = \alpha, \quad j(Q) = \beta.$$

If $\beta \in \mathbb{C}$, then Q is non-cuspidal, so $Q = [\tau]$ for some $\tau \in \mathcal{H}$.

Proof. Define an affine curve $C_N := \{(\alpha, \beta) \in \mathbb{C}^2 \mid F_N(\alpha, \beta) = 0\}$, and let $\overline{C}_N$ be the projective closure of C_N . Then there is a canonical desingularization $g: Y \rightarrow \overline{C}_N$ over $\mathbb{Q}$ of $\overline{C}_N$, i.e., a nonsingular projective

curve Y over $\mathbb{Q}$ and a surjective proper morphism $g : Y \rightarrow \overline{C}_N$ that is an isomorphism except over the singular points of $\overline{C}_N$, and the pair $(Y, g : Y \rightarrow \overline{C}_N)$ is uniquely determined by $\overline{C}_N$ (up to unique isomorphism).

The embedding $i : \mathbb{C}(x, j) \rightarrow \mathbb{C}(j, J)$ induces a surjective morphism $h : X_0(N) \rightarrow Y$. Therefore $g \circ h : X_0(N) \rightarrow \overline{C}_N$ is surjective. It follows from the irreducibility of $F_N(X, Y)$ and the uniqueness of Y that for any $\alpha, \beta \in \mathbb{C}$ with $F_N(\alpha, \beta) = 0$, there exists $\tau \in \mathcal{H}$ such that $\alpha = x(\tau)$ and $\beta = j(\tau)$. $\square$

For a finite target $P = (\alpha, \beta) \in E$, solving $F_N(\alpha, j) = 0$ gives all possible finite j -values of its preimages. A root determines an $\mathrm{SL}_2(\mathbb{Z})$ -orbit in $\mathcal{H}$, which must be lifted to $X_0(N)$ using coset representatives. The equations supply candidates; their images under φ and their equivalence classes must still be checked. Cuspidal preimages are handled separately.

Exact inputs and certified recognition. For the effective algorithm, the input point is assumed to belong to $E(\overline{\mathbb{Q}})$. Algebraic numbers are represented by exact defining data and isolating regions specifying their complex embeddings. The modular parametrization is fixed with the normalization of Section 2.1, and all polynomial relations used for certification are assumed to have been certified as in Section 2.3. For arbitrary complex inputs, the fibre description remains valid as a mathematical procedure, but no terminating exact implementation for unrestricted complex inputs is asserted.

Lemma 3.2 (Recognition from a finite set of possible images). *Let $t \in \overline{\mathbb{Q}}$ be finite, and define*

$$\mathcal{A}_t = \{O_E\} \cup \{(u, v) \in E(\overline{\mathbb{Q}}) : F_N(u, t) = 0\}.$$

This is an effectively computable finite set. For every $Q \in X_0(N)$ with $j(Q) = t$, one has $\varphi(Q) \in \mathcal{A}_t$. If certified enclosures for $\varphi(Q)$ of arbitrarily small diameter are available on $E(\mathbb{C})$, then $\varphi(Q)$ can be identified exactly in $\mathcal{A}_t$ by a procedure that terminates.

Proof. The polynomial $F_N(U, t)$ is not identically zero. Otherwise the minimal polynomial of t over $\mathbb{Q}$ would divide every coefficient of $F_N(U, V)$ as a polynomial in U , contradicting its irreducibility and its dependence on U . Its distinct roots are computable algebraic numbers, and the Weierstrass equation gives at most two values of v for each root u . Thus $\mathcal{A}_t$ is finite and effective. If $\varphi(Q)$ is finite, the identity $F_N(x, j) = 0$ gives $F_N(x(Q), t) = 0$; the remaining possibility is $\varphi(Q) = O_E$.

Remove duplicate points of $\mathcal{A}_t$ by exact algebraic comparison and isolate its distinct elements in pairwise disjoint neighbourhoods on the projective curve $E(\mathbb{C})$. Refine a certified enclosure of $\varphi(Q)$ until it is contained in one of these neighbourhoods. Since the true image is known to belong to $\mathcal{A}_t$, this identifies the exact image. The process terminates because the finite set has positive separation and the enclosure diameters tend to zero. $\square$

In particular, for an algebraic target P , the predicate $\varphi(Q) = P$ is decided by first identifying the algebraic point $\varphi(Q)$ and then comparing algebraic points exactly. It is not decided by a prescribed number of matching decimal digits.

Analytic enclosures. For $t \notin \{0, 1728\}$, a chosen inverse image $\tau_t \in \mathcal{H}$ of t under j can be specified by an isolating disc and validated local inversion; here $j'(\tau_t) \neq 0$. For $t = 1728$ and $t = 0$, take $\tau_t = i$ and $\tau_t = \rho = (-1 + \sqrt{-3})/2$, respectively. Certified inversion and subsequent evaluation must propagate the uncertainties in both t and τ_t .

The Eichler integral admits explicit truncation bounds. Write $f(z) = \sum_{n \geq 1} a_n q^n$, where $q = e^{2\pi iz}$. The Hasse bound and the Hecke relations give $|a_n| \leq \sigma_0(n) \sqrt{n} \leq 2n$, where $\sigma_0(n)$ is the number of positive divisors of n . Consequently, if $|q| \leq r < 1$, then

$$\left| \gamma(z) - \sum_{n=1}^T \frac{a_n}{n} q^n \right| \leq \frac{2r^{T+1}}{1-r}.$$

Together with outward-rounded arithmetic, this gives certified enclosures for the integral at each interior candidate. Certified period computations and elliptic-function evaluation must also be included; the integral's tail bound alone does not bound the error of the full parametrization.

One way to handle O_E without affine poles is to perform the recognition on the torus $\mathbb{C}/\Lambda_E$. Fix a certified period basis $\Lambda_E = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. For each $R \in \mathcal{A}_t$, compute a certified elliptic logarithm λ_R modulo Λ_E , with $\lambda_{O_E} = 0$. Identify the class of $\gamma(z)$ among the finite set $\{\lambda_R \bmod \Lambda_E : R \in \mathcal{A}_t\}$, using enclosures and separation on the torus. Periods and elliptic logarithms can be enclosed using validated

elliptic integrals. Under the real-linear coordinates defined by (ω_1, ω_2) , the comparisons are made in $\mathbb{R}^2/\mathbb{Z}^2$; enclosures crossing a boundary are treated modulo $\mathbb{Z}^2$. Distinct algebraic points have distinct logarithm classes, so this is the same terminating recognition procedure as in Lemma 3.2. Thus the pole predicate is decided by recognition of O_E , rather than by a large affine value of x .

Distinct points in a fibre. For each distinct root t , fix the same lift τ_t for all coset representatives M_n . Let

$$G_t = \text{Stab}_{\text{SL}_2(\mathbb{Z})}(\tau_t).$$

Then

$$[M_n \tau_t] = [M_m \tau_t] \text{ in } X_0(N) \iff M_n s M_m^{-1} \in \Gamma_0(N) \text{ for some } s \in G_t.$$

This is a finite test on integer matrices: membership in $\Gamma_0(N)$ means that the lower-left entry is divisible by N . For $t \notin \{0, 1728\}$, one has $G_t = \{\pm I\}$. For the chosen special lifts, $G_{1728} = \langle S \rangle$ and $G_0 = \langle ST \rangle$, where

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The two groups have orders 4 and 6, respectively. Candidates with distinct j -values are distinct points. Cusps are represented once modulo $\Gamma_0(N)$, using exact cusp equivalence. Points are never identified merely because they have the same pair (j, J) , since different points above a singularity of the plane model must be retained.

Algorithm 2 Computing a complete fibre over an algebraic point

Input: A finite point $P = (\alpha, \beta) \in E(\overline{\mathbb{Q}})$ with its complex embedding specified, certified relations F_N, f_N , and the certified analytic operations described above.

Output: One representative of each distinct point of $\varphi^{-1}(P)$.

- 1: Find the distinct algebraic roots $j_1, \dots, j_{L'}$ of $F_N(\alpha, j) = 0$, and the finite root set Z of $f_N(\alpha, J) = 0$.
 - 2: For each j_i , choose a certified lift $\tau_i \in \mathcal{H}$ with $j(\tau_i) = j_i$. Take $\tau_i = i$ if $j_i = 1728$ and $\tau_i = (-1 + \sqrt{-3})/2$ if $j_i = 0$. For other values, validated inversion of j , or validated periods of $y^2 + xy = x^3 - 36x/(j_i - 1728) - 1/(j_i - 1728)$, give such a lift.
 - 3: Initialize $\text{Inv} = \emptyset$ and construct complete coset representatives $\{M_n : 1 \leq n \leq \mu\}$.
 - 4: For every $Q = [M_n \tau_i]$, identify $\varphi(Q) \in \mathcal{A}_{j_i}$ by Lemma 3.2. Append Q exactly when its identified image equals P . The condition $J(Q) \in Z$ may be used as an exclusion filter only when a certified enclosure is disjoint from Z ; an inconclusive filter does not discard Q .
 - 5: Identify all cusp images by Algorithm 4 and append each cusp class whose image is P .
 - 6: Remove repetitions by the stabilizer test and exact cusp equivalence, and return Inv .
-

These steps give the full fibre under the stated certification requirements. Every non-cuspidal preimage has j -value among the roots found in Step 1, and the coset enumeration covers every point above each root. Finite-image recognition terminates for every candidate. The cusp step supplies the remaining points, and the equivalence test removes precisely repeated representatives.

Remark 3.3. *There exist $\tau_1, \tau_2 \in \mathcal{H}$ with $[\tau_1] \neq [\tau_2]$, $j(\tau_1) = j(\tau_2)$ and $j(N\tau_1) = j(N\tau_2)$. These points correspond to singular points of the plane model of $X_0(N)$ determined by the classical modular equation [15, 22]. For such points $[\tau_1]$ and $[\tau_2]$, we generally have $\varphi([\tau_1]) \neq \varphi([\tau_2])$. Therefore, in the image-recognition step, points of $X_0(N)$ determined by (j, J) may not belong to $\varphi^{-1}(P)$, and we must verify their images under φ to make the final determination.*

Example 3.4. *Consider the elliptic curve 389a1 defined by:*

$$y^2 + y = x^3 + x^2 - 2x. \tag{8}$$

The genus g of $X_0(389)$ is 32, $\mu = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(389)] = 390$, $\deg(\varphi) = 40$, $K = \deg_{F_{389}}(x) = 390 = \mu$, $L = 2 \cdot \deg(\varphi) = 80$. We could factor the polynomial $F_{389}(0, j)$ into two irreducible polynomials $f_1(j)$ and

$f_2(j)$, where

$$\begin{aligned}
f_1(j) &:= j^{40} + \dots + 2^{60} \cdot 3^{12} \cdot 5^{12} \cdot 41^6 \cdot 59^3 \cdot 83^6 \\
&\quad \cdot 25284173^3 \cdot 2266783937522246669759^3 \\
&\quad \cdot 21020023204008338127670158568049294014 \\
&\quad \cdot 9512434767923557515056430506879436133^3 \\
f_2(j) &:= j^{40} + \dots + 2^{59} \cdot 3^{12} \cdot 5^{11} \cdot 11^{12} \cdot 8429^3 \cdot 170801^3 \\
&\quad \cdot 99507788413092762738223^3 \\
&\quad \cdot 7060906424360528718534826263851663910667 \\
&\quad \cdot 9308687516043502683423204467607^3
\end{aligned}$$

The original numerical computation locates a candidate near $\tau \approx -0.02127691418 + 2.268594296 \times 10^{-5}i$. The reported image is

$\varphi(\tau) = (0, 0)$, $j(\tau) \approx -8.747731379 \times 10^{53} - 2.618292441 \times 10^{54}i$ and $j(389\tau) \approx -458.3909478 - 562.7806563i$, with the two invariants numerically matched to roots of $f_2(j)$ and $f_1(j)$ respectively.

For an imaginary quadratic order of discriminant $D < 0$, let $H_D(T)$ be its class polynomial. The following table is an illustrative selection of rank-one optimal curves of conductor at most 100, with one chosen generator of the free part of each listed Mordell–Weil group. It is not used as an exhaustive census of all such curves or all choices of generator. It records the original fibre computations in the monic normalization

$$f_P(T) = \prod_{Q \in \varphi^{-1}(P)} (T - j(Q)).$$

The product is over distinct geometric points of the fibre; coincident j -values still give repeated factors. Since P has infinite order and cusp images are torsion, these fibres contain no cusps. The verification status of these computations is as stated in Section 2.4.

E	Chosen generator P	$f_P(j)$
37a1	(0, 0)	$H_{-7}(j)^2$
43a1	(0, 0)	$H_{-7}(j)^2$
57a1	(2, 1)	$H_{-48}(j)^2$
58a1	(0, 1)	$H_{-20}(j)^2$
61a1	(1, 0)	$H_{-12}(j)^2$
65a1	(1, 0)	$H_{-16}(j)^2$
77a1	(2, 3)	$H_{-35}(j)^2$
79a1	(0, 0)	$H_{-7}(j)^2$
82a1	(0, 0)	$H_{-4}(j)^2$
83a1	(0, 0)	$H_{-19}(j)^2$
88a1	(2, 2)	$H_{-96}(j)^2$
89a1	(0, 0)	$H_{-8}(j)^2$
91a1	(0, 0)	$H_{-27}(j)^4$
91b1	(-1, 3)	$2^{-20} f_{91b1}(j)$
92b1	(1, 1)	$H_{-44}(j)^2$
99a1	(0, 0)	$H_{-8}(j)^4$

For 91b1, the primitive integer polynomial is

$$\begin{aligned}
f_{91b1}(j) &:= 2^{15} \cdot 5^{11} \cdot 991^3 \cdot 16572269011^3 \\
&\quad - 2^9 \cdot 3^4 \cdot 5^{10} \cdot 23 \cdot 40341091849 \cdot 788088043784206924489867 j \\
&\quad + 5^6 \cdot 248698909 \cdot 269818358221989089513089057757 j^2 \\
&\quad - 2^7 \cdot 3^4 \cdot 5^4 \cdot 49633 \cdot 5413524016643 j^3 \\
&\quad + 1048576 j^4.
\end{aligned}$$

Remark 3.5. The curve 91b1 is

$$y^2 + y = x^3 + x^2 - 7x + 5,$$

and $(-1, 3)$ is the chosen generator of infinite order. Exact polynomial arithmetic verifies that f_{91b1} has content one, is irreducible over $\mathbb{Q}$, and has leading coefficient 2^{20} ; see `verify91b1_polynomial.gp` in [26]. Consequently, the monic minimal polynomial of each of its roots is $2^{-20}f_{91b1}$, and none of these roots is an algebraic integer. Applied to the reported fibre, this proves that its j -values are not singular moduli. Schneider's theorem then implies that the corresponding representatives in $\mathcal{H}$ are transcendental. The other rows record CM fibres for their chosen generators. The occurrence of CM and non-CM fibres in this sample suggests an experimental question about the arithmetic of generator fibres; the table alone gives no general classification.

4 Algorithm for Computing Poles of φ

For any $\tau \in \mathcal{H}$, we have

$$x(\tau) = \infty \iff \wp(\gamma(\tau)) = \infty \iff \gamma(\tau) \in \Lambda_E,$$

where Λ_E is the period lattice of E . Both coordinate functions have pole support $\varphi^{-1}(O_E)$. In particular, every zero of γ represents a point of this fibre.

Let $x(\tau_0) = \infty$ for some $\tau_0 \in \mathcal{H}$. Then there exists a punctured neighborhood U of τ_0 such that for all $\tau \in U$, $x(\tau) \neq \infty$ and

$$F_N(x(\tau), j(\tau)) = \sum_{k=0}^K A_k(j(\tau)) x(\tau)^k \equiv 0.$$

This implies

$$A_K(j) + \frac{A_{K-1}(j)}{x} + \dots + \frac{A_0(j)}{x^K} = 0.$$

Taking the limit as $\tau \rightarrow \tau_0$ (and thus $x \rightarrow \infty$) yields $A_K(j(\tau_0)) = 0$, showing that $j(\tau_0)$ is a zero of $A_K(j) = 0$. Then we have the following algorithm.

Algorithm 3 Non-cuspidal points in the fibre over O_E

Input: Modular polynomial $F_N(x, j)$.

Output: Distinct non-cuspidal points of $\varphi^{-1}(O_E)$.

- 1: Construct right coset representatives $\{M_n | 1 \leq n \leq \mu\}$ of $\Gamma_0(N)$ in $SL_2(\mathbb{Z})$ and set $S := \emptyset$.
 - 2: Find zeros $j_1, \dots, j_{L'}$ of $A_K(j) = 0$.
 - 3: For each j_k , find z_k such that $j(z_k) = j_k$ using the certified lift construction in Algorithm 2.
 - 4: For every $Q = [M_n z_k]$, identify $\varphi(Q) \in \mathcal{A}_{j_k}$ by Lemma 3.2. Append Q exactly when the identified image is O_E .
 - 5: Remove repetitions by the stabilizer test of Section 3 and return S .
-

The roots of A_K supply every possible finite j -value of a point mapping to O_E . The finite enumeration and image recognition therefore give the complete non-cuspidal part of this fibre. Add the cusps mapping to O_E , as determined in Section 5, to obtain the full fibre.

Proposition 4.1 (Non-CM zeros of the Eichler integral for 38a1). *Let*

$$E: \quad y^2 + xy + y = x^3 + 9x + 90$$

be the optimal elliptic curve 38a1, and let γ be the Eichler integral of its associated normalized newform, with the normalization of Section 2. Put

$$\begin{aligned} h(T) = & T^4 - 4534998854T^3 + 39952945838749259709T^2 \\ & - 5295752119436627969393180T + 256453780788797510341879944067. \end{aligned}$$

For every root j_0 of h , there exists $\tau \in \mathcal{H}$ such that

$$\gamma(\tau) = 0, \quad j(\tau) = j_0.$$

None of the roots of h is a singular modulus. Consequently, γ has zeros representing non-CM points of $X_0(38)$.

Proof. We first record the exact polynomial data used in the argument. The relation $F_{38}(X, T)$ has bidegree $(60, 12)$, and

$$[X^{60}]F_{38}(X, T) = h(T)^2.$$

Its coefficients, viewed as polynomials in T , have greatest common divisor 1, and $F_{38}(X, 8)$ modulo 13 is irreducible of degree 60. These checks imply that F_{38} is irreducible over $\mathbb{Q}$. The modular degree is $d = 6$ and $\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(38)] = 60$. Thus the bound in Lemma 2.5 is

$$B_F = 2 \cdot 6 \cdot 60 + 60 \cdot 12 = 1440.$$

Exact substitution of the q -expansions gives

$$F_{38}(x(q), j(q)) = O(q^{1473}),$$

which certifies the identity by that lemma. Together with irreducibility, this identifies F_{38} as the primitive irreducible relation. The accompanying PARI/GP script `verify38_certificate.gp` reproduces these checks.

To show that h has no real root, set

$$\begin{aligned} a &= 34811392187303931380, \\ b &= 5295752119436627969393180, \\ c &= 256453780788797510341879944067. \end{aligned}$$

A direct polynomial identity gives

$$h(T) = T^2(T - 2267499427)^2 + aT^2 - bT + c.$$

Here $a > 0$, and

$$4ac - b^2 = 7665062053305265651119067356625678860036993977440 > 0.$$

Consequently,

$$aT^2 - bT + c = a \left(T - \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a} > 0 \quad (T \in \mathbb{R}),$$

so $h(T) > 0$ for every real T .

Suppose that a root of h were a singular modulus of discriminant $D < 0$. Its minimal polynomial over $\mathbb{Q}$ is the class polynomial $H_D(T)$, by the first main theorem of complex multiplication; see, for example, [25, Corollary 21.2]. Hence H_D would divide h . However, the root of H_D corresponding to the principal ideal class is real. Indeed, a representative of that class has real part 0 or $-1/2$, so its q -parameter is real and its j -value is real. This would give a real root of h , a contradiction.

Fix a root j_0 of h . Let C be the closure of the image of (x, j) in $\mathbb{P}^1 \times \mathbb{P}^1$. The certified irreducible relation F_{38} defines its affine part. Homogenizing in X shows that

$$(\infty, j_0) \in C,$$

because the leading coefficient is $h(j_0)^2 = 0$. The morphism

$$(x, j) : X_0(38) \longrightarrow C$$

is surjective, since its image is closed and contains the dense affine image. Therefore there exists $Q \in X_0(38)$ with

$$x(Q) = \infty, \quad j(Q) = j_0.$$

As j_0 is finite, Q is non-cuspidal. Choose a representative $\tau_0 \in \mathcal{H}$ of Q . If Λ_E denotes the period lattice of E , then

$$\varphi(Q) = O_E, \quad \gamma(\tau_0) = \lambda \in \Lambda_E.$$

For $M \in \Gamma_0(38)$, put

$$P_f(M) = \gamma(Mz) - \gamma(z), \quad z \in \mathcal{H}.$$

The invariance of $f(z) dz$ shows that this difference is independent of z , and

$$P_f(M_1 M_2) = P_f(M_1) + P_f(M_2).$$

Thus the image of P_f is an additive subgroup of Λ_E . By the period-lattice equality (5), this image generates Λ_E , and hence

$$P_f(\Gamma_0(38)) = \Lambda_E.$$

Choose $M \in \Gamma_0(38)$ with $P_f(M) = -\lambda$, and set $\tau = M\tau_0$. Then

$$\gamma(\tau) = \gamma(\tau_0) + P_f(M) = 0, \quad j(\tau) = j(\tau_0) = j_0.$$

Since j_0 is not a singular modulus, τ is not a CM point. $\square$

Remark 4.2. *This proposition answers negatively the universal question in Kodgis [16, Section 5, p. 10]. Peluse's sufficient conditions for CM zeros and her account of Ono's speculation about generic behaviour [19] have different scopes. The proof above does not depend on assigning an exact zero or period to a decimal approximation of a particular representative.*

5 Algorithm for Computing Exact Values of φ at Cusps

Let $[\tau_0]$ be a cusp of $X_0(N)$ with $x(\tau_0) \neq \infty$. Then $j(\tau_0) = \infty$ and there exists a punctured neighborhood U of τ_0 such that for all $\tau \in U$, $j(\tau) \neq \infty$ and

$$F_N(x(\tau), j(\tau)) = \sum_{l=0}^L B_l(x(\tau)) j(\tau)^l \equiv 0.$$

This implies

$$B_L(x) + \frac{B_{L-1}(x)}{j(\tau)} + \cdots + \frac{B_0(x)}{j(\tau)^L} \equiv 0.$$

Taking the limit as $\tau \rightarrow \tau_0$ (so $j(\tau) \rightarrow \infty$) yields

$$B_L(x(\tau_0)) = 0.$$

Thus every finite cusp value of x is a root of $B_L(x)$. Conversely, the roots of $B_L(x)$ describe the finite points of the plane curve $F_N(x, j) = 0$ lying above $j = \infty$ in its closure in $\mathbb{P}^1 \times \mathbb{P}^1$. Lifting first to the normalization of this relation curve and then to $X_0(N)$ shows that they come from cusps. Hence the finite cusp values are precisely the roots of $B_L(x)$:

$$\{x([\tau]) : \tau \in \mathbb{Q} \cup \{i\infty\}, x([\tau]) \neq \infty\} = \{\alpha \in \mathbb{C} : B_L(\alpha) = 0\}.$$

Remark 5.1. *If the equation $B_L(x) = 0$ has no solutions, this indicates that all cusps of $X_0(N)$ map to O_E under φ .*

We combine the leading-coefficient equations with certified recognition in their finite algebraic candidate set. For $M \in \Gamma_0(N)$, fix $z_* \in \mathcal{H}$ and put $P_f(M) = \gamma(Mz_*) - \gamma(z_*)$. This period is independent of z_* ; both terms admit the explicit interior error bound from Section 3. Cusp representatives can be chosen as s/r , with $r \mid N$. A prime $p \equiv 1 \pmod{N}$ satisfies the cusp-equivalence conditions below, so the prime search terminates. The Hasse bound gives $p + 1 - a_p > 0$.

Algorithm 4 Computation of the exact values of φ at cusps

Input: modular polynomial $F_N(x, j)$ and $G_N(y, j)$.

Output: Exact values of φ at cusps.

- 1: Initialize an empty vector $Z \leftarrow \emptyset$; for each root x_i of $B_L(x) = 0$ and each root y_j of $C(y) = 0$, if (x_i, y_j) satisfies equation (2) $Z \leftarrow Z \cup \{(x_i, y_j)\}$, where $C(y)$ is the leading coefficient of $G_N(y, j)$ as a polynomial in j .
- 2: Adjoin O_E to Z and remove equal algebraic points. Every cusp image belongs to this finite set.
- 3: For each cusp $[\frac{s}{r}]$ of $X_0(N)$, compute certified enclosures of $\varphi([\frac{s}{r}])$ by the following step.
 - (1) Find minimal prime p with $(p, N) = 1$ such that $[\frac{s}{r}] = [\frac{ps}{r}] = [\frac{rj+s}{pr}]$ for all $0 \leq j \leq p-1$, where we use the criterion from Proposition 2.2.3 in [10] to decide whether or not two cusps are equivalent.
 - (2) For each $j = 0, 1, \dots, p$, construct $M_j \in \Gamma_0(N)$ such that

$$M_p \left(\frac{ps}{r} \right) = M_j \left(\frac{rj+s}{pr} \right) = \frac{s}{r}$$

by the method contained in the proof of Proposition 2.2.3 in [10].

- (3) For $j = 0, 1, \dots, p$, compute the period $P_f(M_j)$ using the method from Proposition 2.10.3 in [10].

- (4) Compute

$$\gamma \left(\frac{s}{r} \right) := \frac{1}{p+1-a_p} \sum_{j=0}^p P_f(M_j)$$

to a given precision, where a_p is the coefficient of q^p in the q -expansion of f . Recall that f is the newform for $\Gamma_0(N)$ associated to E by φ .

- (5) Propagate the period error bounds to certified enclosures of the cusp image, using elliptic integrals or the projective curve. Refine until the image is uniquely identified in Z , as in Lemma 3.2. The torus $\mathbb{C}/\Lambda_E$ treats O_E without affine poles.

- 4: Output every cusp class with its exactly identified image.
-

Remark 5.2. The period evaluation uses the Hecke relation and cusp-equivalence computations described in [10, 11]. Finite separation supplies the exact recognition step.

Example 5.3. Consider $N = 176$. The elliptic curve 176a1 is given by

$$y^2 = x^3 - 4x - 4.$$

The leading coefficient $B_L(x)$ in $F_{176}(x, j)$ as a polynomial in j is 1. Since $B_L(x) = 0$ has no solutions, this shows that φ maps all cusps to O_E . The leading coefficient $A_K(j)$ of x in $F_{176}(x, j)$ as a polynomial in x is 1. Consequently, the equation $A_K(j) = 0$ has no solutions, which indicates that no points in $\mathcal{H}$ map to O_E under φ .

Thus, the fibre over O_E consists entirely of cusps, i.e.,

$$\varphi^{-1}(O_E) = \{[1], [\frac{1}{2}], [\frac{1}{4}], [\frac{3}{4}], [\frac{1}{8}], [\frac{1}{11}], [\frac{1}{16}], [\frac{1}{22}], [\frac{1}{44}], [\frac{3}{44}], [\frac{1}{88}], [\frac{1}{176}]\}.$$

The modular parametrization $\varphi : X_0(176) \rightarrow E$ has degree 16. However, $\varphi^{-1}(O_E)$ contains only 12 distinct points. Using the criterion of Corbett and Saha (Theorem 1.1, [9]), we find that φ is ramified at the cusps $[\frac{1}{4}]$, $[\frac{3}{4}]$, $[\frac{1}{44}]$, $[\frac{3}{44}]$ and each has ramification index 2. Thus when counting with multiplicity, the fiber satisfies:

$$\sum_{[\tau] \in \varphi^{-1}(O_E)} e_\varphi([\tau]) = 16,$$

where $e_\varphi([\tau])$ denotes the ramification index at $[\tau]$.

6 Algorithm for Finding Branch Values of φ

Let

$$Q = [\tau_0] \in X_0(N)$$

be a non-cuspidal ramification point of

$$\varphi : X_0(N) \longrightarrow E,$$

with $\varphi(Q) \neq O_E$, where O_E is the point at infinity of E , and write

$$\varphi(Q) = (\alpha, \beta) \in E.$$

Let

$$x = \varphi^* x_E = x_E \circ \varphi \in \mathbb{C}(X_0(N))$$

be the pullback of the Weierstrass x -coordinate. Since Q is a ramification point of φ , we have

$$e_\varphi(Q) > 1.$$

Hence the local degree of the composed map

$$x : X_0(N) \longrightarrow \mathbb{P}^1$$

at Q satisfies

$$e_x(Q) = e_{x_E}(\varphi(Q)) e_\varphi(Q) > 1.$$

Thus

$$\alpha = x(Q)$$

is a branch value of the meromorphic function x .

First assume that $j(Q) \notin \{0, 1728\}$. Since the map $j : X_0(N) \rightarrow X(1)$ is unramified at Q , we may choose a sufficiently small coordinate neighbourhood U of Q on which j is injective.

Let

$$e = e_x(Q).$$

Choose a local parameter u at Q . Since x has local degree e at Q , after replacing u by another local parameter if necessary, we may write

$$x - \alpha = u^e.$$

Since Q is non-cuspidal, the function j is holomorphic at Q . Therefore

$$j(u) = j_0 + b_1 u + b_2 u^2 + \cdots, \quad j_0 = j(Q) = j(\tau_0), \quad b_1 \neq 0.$$

Now let z be a value of x close to α , with $z \neq \alpha$. Locally near Q , the equation

$$x = z$$

is equivalent to

$$u^e = z - \alpha.$$

After the ramified change of variable

$$z - \alpha = t^e,$$

the e local sheets above z are given by

$$u = \zeta_e^i t, \quad i = 0, \dots, e-1,$$

where ζ_e is a primitive e -th root of unity. Along these local sheets, the corresponding j -values are

$$j_i(t) = j(\zeta_e^i t), \quad i = 0, \dots, e-1.$$

In particular,

$$j_i(t) \longrightarrow j_0 \quad \text{as } t \rightarrow 0.$$

For $z \neq \alpha$ sufficiently close to α , these local sheets are among the sheets described by the algebraic relation

$$F_N(z, j) = 0.$$

Thus $F_N(z, j)$, as a polynomial in j , has roots given locally by the corresponding values $j_i(t)$. Since j is injective on U , these e values are pairwise distinct for sufficiently small $t \neq 0$. Since the e sheets above z coalesce to the single point Q as $z \rightarrow \alpha$, the corresponding e roots tend to the same value j_0 . Hence j_0 is a multiple root of

$$F_N(\alpha, j).$$

Consequently,

$$F_N(\alpha, j_0) = 0, \quad \frac{\partial F_N}{\partial j}(\alpha, j_0) = 0.$$

Therefore $(\alpha, j(\tau_0))$ is a common zero of

$$F_N(x, j) \quad \text{and} \quad \frac{\partial F_N}{\partial j}(x, j).$$

Consequently, under $j(Q) \notin \{0, 1728\}$, the x -coordinate α is a zero of

$$\text{Res}_j \left(F_N(x, j), \frac{\partial F_N}{\partial j}(x, j) \right).$$

This condition is only necessary under the stated assumption. The resultant may also vanish because of singularities of the plane model, leading-coefficient degenerations, coincidences among distinct points of $X_0(N)$, or ramification of the coordinate map x_E itself. Since F_N is irreducible and the characteristic is zero, this resultant is not identically zero. Thus its zeros form a finite set of candidates, and the genuine branch values are selected by the subsequent fibre verification.

The same argument applies with x replaced by y , or with j replaced by J , provided that the auxiliary function is unramified at Q . Since $J = j \circ W_N$, where W_N is the Fricke involution, J is unramified at every non-cuspidal point outside $J^{-1}(\{0, 1728\})$. To cover the exceptional cases, we separately check the images of the finite set

$$\mathcal{S} = j^{-1}(\{0, 1728\}) \cup J^{-1}(\{0, 1728\}) \subset X_0(N),$$

without requiring these images to satisfy any of the resultant conditions. The fibre over O_E is also checked separately, using Algorithm 3 for its non-cuspidal points and Algorithm 4 for its cuspidal points. All fibre cardinalities used below refer to complete fibres obtained by the certified procedure above and counted as distinct points of $X_0(N)$, including cusps. Since $\deg(\varphi) = d$, one has

$$\sum_{Q \in \varphi^{-1}(P)} e_\varphi(Q) = d, \quad \sum_{Q \in \varphi^{-1}(P)} (e_\varphi(Q) - 1) = d - \#\varphi^{-1}(P).$$

Consequently, the test $\#\varphi^{-1}(P) < d$ is valid only after completeness and removal of repeated representatives have been established. Images of the exceptional set and of the cusps are identified by the same finite-candidate recognition procedure and compared as algebraic points of E .

Writing $d = \deg \varphi$, we use

$$P \text{ is a branch value of } \varphi \iff \#\varphi^{-1}(P) < d.$$

Thus we have the following algorithm.

Algorithm 5 Algorithm for Finding Branch Values of φ

Input: $F_N(x, j)$, $f_N(x, J)$, $G_N(y, j)$ and $g_N(y, J)$.

Output: Set of branch values of φ .

1: Compute

$$R_1(x) = \text{Resultant}(F_N(x, j), \frac{\partial F_N}{\partial j}(x, j), j),$$

$$R_2(x) = \text{Resultant}(f_N(x, J), \frac{\partial f_N}{\partial J}(x, J), J),$$

$$U_0(x) = \gcd(R_1(x), R_2(x)) \text{ and } U(x) = \frac{U_0(x)}{\gcd(U_0(x), U'_0(x))}.$$

2: Compute

$$R_3(y) = \text{Resultant}(G_N(y, j), \frac{\partial G_N}{\partial j}(y, j), j)$$

$$R_4(y) = \text{Resultant}(g_N(y, J), \frac{\partial g_N}{\partial J}(y, J), J)$$

$$V_0(y) = \gcd(R_3(y), R_4(y)) \text{ and } V(y) = \frac{V_0(y)}{\gcd(V_0(y), V'_0(y))}.$$

3: Find all (x_i, y_j) with $(x_i, y_j) \in E$ for each solution x_i of $U(x) = 0$ and y_j of $V(y) = 0$; then compute $\varphi^{-1}((x_i, y_j))$ using Algorithm 2 and decide whether (x_i, y_j) is a branch value. This step detects every finite branch value having a non-cuspidal ramification point outside $\mathcal{S}$.

4: Let $\mathcal{C}$ be the set of cusps of $X_0(N)$. Independently of Steps 1–3, test every point of

$$\varphi(\mathcal{S}) \cup \varphi(\mathcal{C}) \cup \{O_E\}$$

by computing its full fibre and retaining it if the fibre contains fewer than d distinct points. Use Algorithm 2 for finite points, and Algorithms 3 and 4 for O_E .

5: Output the union of the results from Steps 3 and 4.

Example 6.1. Consider the elliptic curve 46a1 defined by

$$y^2 + xy = x^3 - x^2 - 10x - 12.$$

The genus g of $X_0(46)$ is 5, $\mu = [SL_2(\mathbb{Z}) : \Gamma_0(46)] = 72$, $\deg(\varphi) = 5$, $K = \deg_x F_{46}(x, j) = 72 = \mu$, and $L = \deg_j F_{46}(x, j) = 2 \deg(\varphi) = 10$.

Step 1. The square-free part of the greatest common divisor of

$$\text{Res}_j \left(F_{46}(x, j), \frac{\partial F_{46}}{\partial j}(x, j) \right)$$

and

$$\text{Res}_J \left(f_{46}(x, J), \frac{\partial f_{46}}{\partial J}(x, J) \right)$$

factors as $U = U_1 U_2 \cdots U_{14}$, where

$$U_1 = -4 + x,$$

$$U_2 = 4 + 3x + x^2,$$

$$U_3 = 3184 + 2472x + 567x^2 - 70x^3 + 23x^4.$$

Step 2. The square-free part of the greatest common divisor of

$$\text{Res}_j \left(G_{46}(y, j), \frac{\partial G_{46}}{\partial j}(y, j) \right)$$

and

$$\text{Res}_J \left(g_{46}(y, J), \frac{\partial g_{46}}{\partial J}(y, J) \right)$$

factors as $V = V_1 V_2 V_3$, where

$$\begin{aligned} V_1 &= 2 + y, \\ V_2 &= 1996 - 1004y + 711y^2 + 14y^3 + 27y^4, \\ V_3 &= 7416293824 + 9784853696y - 1586824496y^2 \\ &\quad - 3448946432y^3 + 971389940y^4 + 12153116y^5 \\ &\quad + 7085747y^6 + 37030y^7 + 12167y^8. \end{aligned}$$

Step 3. The candidate elimination is summarized in Table 4. The descriptions of the fibres in this table refer to the numerical computations; they are separated from the exact polynomial candidate conditions. The two quartic factors associated with V_2 are

y -factor	Compatible x -condition	Outcome of the original computation
$V_1 = y + 2$	$U_1 U_2 = 0$	Three points, reported as unramified fibres. The fibre of $(4, -2)$ contains $[1]$ and $[1/23]$.
V_2	Two quartic factors below	No common factor with U in the reported factorization; discarded at the resultant stage.
V_3	$U_3 = 0$	Eight distinct points on E , numerically identified as branch values.

Table 4: Candidate elimination for 46a1. Exceptional images are handled independently by Algorithm 5.

$$(1873 + 160x - 234x^2 - 16x^3 + 9x^4)(112 + 40x - 57x^2 - 10x^3 + 9x^4).$$

The eight remaining points form

$$\mathcal{B} = \{(x, y) \in E : U_3(x) = 0, V_3(y) = 0\}.$$

This revisits the branch-value example of Delaunay [11, p. 121]. The present example illustrates how they arise from the coordinate relations and their resultants.

The V_1 row also illustrates why plane-coordinate coincidences must not be counted as ramification of φ . Put $x_1 = (-3 - i\sqrt{7})/2$, $x_2 = (-3 + i\sqrt{7})/2$. The three points in that row are $(4, -2), (x_1, -2), (x_2, -2)$.

Let $\tau_0 = \frac{7+\sqrt{-7}}{4}$. Define the following matrices and put $\tau_k = M_k \tau_0$ for $1 \leq k \leq 4$:

$$\begin{aligned} M_1 &= \begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix}, & M_2 &= \begin{pmatrix} 0 & -1 \\ 1 & 5 \end{pmatrix}, \\ M_3 &= \begin{pmatrix} 1 & 14 \\ 2 & 29 \end{pmatrix}, & M_4 &= \begin{pmatrix} 1 & 16 \\ 2 & 33 \end{pmatrix}. \end{aligned}$$

These represent four distinct points on $X_0(46)$, but

$$j(\tau_k) = J(\tau_k) = -3375 \quad (1 \leq k \leq 4).$$

Thus they lie above the same singular point $(-3375, -3375)$ of $Z_0(46)$ (see Section 10). The original numerical image computation gives the following pairings:

$$\begin{aligned} [\tau_1] &\in \varphi^{-1}((x_1, -2)), & [\tau_2] &\in \varphi^{-1}((x_1, y_1)), \\ [\tau_3] &\in \varphi^{-1}((x_2, y_2)), & [\tau_4] &\in \varphi^{-1}((x_2, -2)), \end{aligned}$$

where

$$y_1 = \frac{7 + i\sqrt{7}}{2}, \quad y_2 = \frac{7 - i\sqrt{7}}{2}.$$

The example shows that although $j = -3375$ is a double root of $F_{46}(x_1, j)$, there exist two distinct points $[\tau_1]$ and $[\tau_2]$ on $X_0(46)$ satisfying

$$x(\tau_1) = x(\tau_2) = x_1, \quad j(\tau_1) = j(\tau_2) = -3375.$$

Step 4. Independently of the resultant conditions, the algorithm tests

$$\varphi(j^{-1}(\{0, 1728\}) \cup J^{-1}(\{0, 1728\})) \cup \varphi(\mathcal{C}) \cup \{O_E\},$$

where $\mathcal{C}$ is the cusp set. The Riemann–Hurwitz formula gives

$$\sum_{Q \in X_0(46)} (e_\varphi(Q) - 1) = 2g(X_0(46)) - 2 = 8.$$

Once each of the eight distinct points of $\mathcal{B}$ has been certified to be a branch value by a complete fibre calculation, these eight fibres account for the entire sum. It then follows that each has exactly one point of ramification index 2, four distinct preimages in total, and that there are no other branch values. This is a conditional completeness check until the individual fibre certificates are supplied; the total count does not replace them. Since 46 is square-free, the parametrization is unramified at all cusps [6].

7 A semi-trace computation for a CM point

We illustrate the computation of a semi-trace of the image of a CM point. In this example the relevant ring class group describes the Galois action explicitly. In addition to the construction below, we give an independent algebraic certificate for the resulting semi-trace.

Example 7.1. Let $E = 389a1$ be the curve in (8). Put $\tau_0 = (-1 + \sqrt{-3})/2$, $\tau_1 = 389\tau_0$, and

$$\Lambda = \mathbb{Z} + \mathbb{Z}\tau_0, \quad \Lambda' = \mathbb{Z}/389 + \mathbb{Z}\tau_0.$$

The cyclic isogeny $z = (\text{id} : \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda')$ represents $[\tau_0] \in X_0(389)$. Its two invariants are $j(\tau_0) = 0$ and $J(\tau_0) = j(\tau_1)$.

Put

$$K = \mathbb{Q}(\sqrt{-3}), \quad \mathcal{O} = \mathbb{Z} + 389\mathcal{O}_K, \quad D = \text{disc}(\mathcal{O}) = -453963.$$

The field $L = K(j(\tau_1))$ is the ring class field of $\mathcal{O}$. Since 389 is inert in K , the ring class group is

$$\text{Pic}(\mathcal{O}) \cong \mathbb{F}_{389^2}^\times / (\mathbb{F}_{389}^\times \text{im}(\mathcal{O}_K^\times)) \cong C_{130}.$$

Here K has class number one, and the image of $\mathcal{O}_K^\times$ in $\mathbb{F}_{389^2}^\times / \mathbb{F}_{389}^\times$ has order 3. Consequently, $[L : K] = 130$ and $[L : \mathbb{Q}] = 260$. The dual of the isogeny represented by z is the unique ascending 389-isogeny from a curve with endomorphism ring $\mathcal{O}$ to one with endomorphism ring $\mathcal{O}_K$. This uniqueness shows that z is defined over L . Since φ is defined over $\mathbb{Q}$,

$$P = \varphi(z) = (x_1, y_1) \in E(L).$$

Moreover, complex conjugation fixes P , since $-\overline{\tau_0} = \tau_0 + 1$.

Step 1. For the modular interpretation below we use the supplied polynomial F_{389} as the coordinate relation defined in Section 2. Its full finite q -expansion certificate is a separate input requirement; the compact verifier here checks its specialization and the subsequent algebraic and analytic certificates. The exact specialization has the form

$$F_{389}(X, 0) = c_0 f_0(X)^3, \quad c_0 \in \mathbb{Q}^\times,$$

where $f_0 \in \mathbb{Z}[X]$ is primitive and irreducible of degree 130. Together with the certified relation $F_{389}(x, j) = 0$, this identifies f_0 as the minimal polynomial of x_1 . Factoring the resultant with the Weierstrass equation gives two irreducible polynomials of degree 130 in the second coordinate. We select g_0 using the certified enclosures

$$\frac{52794735}{1000} < x_1 < \frac{52794736}{1000}, \quad 12130816 < y_1 < 12130817.$$

Exact Sturm counts give one root of f_0 in the first interval and one root of g_0 in the second. The coordinate pairing is certified by the polynomial identities in Proposition 7.2. The enclosures and the algebraic checks are reproduced by `verify389_certificate.gp`.

Let $a > 0$ be the leading coefficient of f_0 . Here and below, a primitive integer minimal polynomial is distinguished from its monic rational normalization f_0/a .

Step 2. Let $G_0 = \text{Gal}(L/K)$, and let c denote complex conjugation. Then

$$G = \text{Gal}(L/\mathbb{Q}) = G_0 \rtimes \langle c \rangle, \quad G_0 \cong C_{130}, \quad c\sigma c = \sigma^{-1} \quad (\sigma \in G_0).$$

For example, the reduced primitive form

$$[273, 111, 427]$$

represents a generator of $\text{Pic}(\mathcal{O})$. The accompanying script enumerates all 130 reduced primitive forms and verifies that this class has order 130. We fix the Artin convention

$$j_{\mathfrak{b}}^{\text{Art}(\mathfrak{a})} = j_{\mathfrak{a}^{-1}\mathfrak{b}},$$

and write σ for the image of the chosen generator. Define

$$P_i = \sigma^{i-1}(P) = (x_i, y_i), \quad J_i = \sigma^{i-1}(j(\tau_1)), \quad 1 \leq i \leq 130.$$

These definitions fix the pairing, and give

$$f_{389}(x_i, J_i) = 0, \quad \sigma^r(P_i) = P_{i+r},$$

where indices are read modulo 130. In particular, the polynomial in this pairing is $f_{389}(x, J)$. The subgroup and semi-trace are

$$H = G_0^2 = \langle \sigma^2 \rangle, \quad |H| = 65, \quad S_H = \sum_{r=0}^{64} P_{2r+1}.$$

Replacing the Artin convention by its inverse does not change H or this sum.

Since x_1 is real and has degree 130 over $\mathbb{Q}$,

$$[K(x_1) : \mathbb{Q}] = 260.$$

Together with $K(x_1) \subseteq L$, this proves

$$L = K(x_1).$$

This field equality uses the degree of the specialized minimal polynomial and the field of definition of P .

Both cusps of $X_0(389)$ are rational and have torsion images. The exact counts $\#E(\mathbb{F}_5) = 9$ and $\#E(\mathbb{F}_7) = 13$ give $E(\mathbb{Q})_{\text{tors}} = 0$, so both images are O_E . In particular, $\gamma(0) \in \Lambda_E$, and Proposition 8.1 gives $F_{389} = f_{389}$. Thus using $F_{389}(x_i, J_i)$ instead of $f_{389}(x_i, J_i)$ in this particular example is justified by Fricke symmetry.

Step 3. For the original construction, one can represent all conjugate points in a single power basis of L . Appendix A fixes the generator $\theta = ax_1 - \sqrt{-3}$, proves that it generates L , and records the trace symmetry used in the recovery. The following certificate avoids publishing the large individual coordinate representations.

Step 4. PARI/GP supports exact elliptic-curve group operations with number-field coordinates, including coordinates represented by polynomial residues. For this large computation we used a separate implementation of the group law to reduce divisions and coefficient growth. An independent certificate of the final semi-trace is given in Proposition 7.2. Its verification uses polynomials over $\mathbb{Q}(\sqrt{389})$ and does not require the individual power-basis representations of all 130 conjugates.

Proposition 7.2 (A compact certificate for the semi-trace). Assume that the supplied F_{389} is the coordinate relation from Section 2, and use the point and subgroup fixed above. Let

$$F = \mathbb{Q}(\sqrt{389}), \quad C(T) = T^3 + T^2 - 2T,$$

where the real embedding is chosen by $\sqrt{389} > 0$. There are polynomials $u, A, B \in F[T]$, supplied in `verify389_certificate.gp`, with

$$\deg u = 65, \quad \deg A = 33, \quad \deg B = 31, \quad u \text{ and } A \text{ monic},$$

which certify

$$S_H = (x_H, y_H),$$

where

$$x_H = \frac{83054760289201}{68231958132816}, \quad y_H = -\frac{1}{2} + \frac{591325257523727551823}{11116205148463643556288} \sqrt{389}.$$

Proof. The nontrivial genus character of $\text{Pic}(\mathcal{O})$ is

$$[\mathfrak{a}] \mapsto \left(\frac{389}{N \mathfrak{a}} \right).$$

It is trivial on principal ideals defining the ring class group: a generator that is scalar modulo 389 has square norm modulo 389. It is nontrivial on the class of an ideal of norm 3. By quadratic reciprocity and Artin reciprocity, its corresponding quadratic extension of K is $K(\sqrt{389})$. Thus

$$L^H = K(\sqrt{389}).$$

Since $cP = P$, the semi-trace is fixed by c , and hence belongs to $E(F)$.

The polynomial u is the factor of f_0 over F selected by the certified interval for x_1 . The exact checks are

$$u(T)\bar{u}(T) = \frac{f_0(T)}{a},$$

and opposite signs at the two endpoints of that interval in the embedding $\sqrt{389} > 0$. Here the bar denotes the nontrivial automorphism of $F/\mathbb{Q}$. Irreducibility of f_0 over $\mathbb{Q}$ and $\deg u = 65$ imply that u is irreducible over F . Since $\mathbb{Q}(x_1) = L^{(c)}$, its roots are exactly the x -coordinates of the 65 points in the H -orbit of P .

Write the polynomial remainder of $g_0(Y)$ modulo $Y^2 + Y - C(T)$ as

$$g_0(Y) \equiv R_0(T) + R_1(T)Y \pmod{Y^2 + Y - C(T)}.$$

The certificate verifies

$$\gcd(R_1, f_0) = 1, \quad u \mid R_0B - R_1A.$$

Since u is irreducible and $\deg B < \deg u$, we have $\gcd(B, u) = 1$. Together with the selected factor g_0 , this fixes the second coordinate as $-A/B$ at each root of u . In particular, it fixes the pairing of the two coordinates.

The remaining exact identities are

$$A(T)^2 - A(T)B(T) - B(T)^2C(T) = u(T)(T - x_H),$$

$$A(x_H) = (y_H + 1)B(x_H), \quad u(x_H)B(x_H) \neq 0.$$

On E , let

$$\ell = A(x_E) + B(x_E)y_E.$$

Its conjugate under the elliptic involution is $A(x_E) - B(x_E) - B(x_E)y_E$, and their product is $u(x_E)(x_E - x_H)$. The preceding identities therefore give

$$\text{div}(\ell) = \sum_{\eta \in H} (\eta P) + (-(x_H, y_H)) - 66(O_E).$$

There is a pole of order 66 at O_E , because $\deg A = 33$, whereas $\deg B = 31$ gives pole order 65 for $B(x_E)y_E$. Passing to $\text{Pic}^0(E) \cong E$ yields

$$\sum_{\eta \in H} \eta P = (x_H, y_H).$$

All the stated polynomial identities and root-isolation checks are verified by the accompanying script. The identity $F_{389}(x, j) = 0$ is used from the preceding construction of the modular polynomial. $\square$

Remark 7.3. *The field and the conjugation symmetry of S_H can also be explained before inspecting its coordinates. Let $Q_1, \dots, Q_{130}$ be the points of $X_0(389)$ above $j = 0$. Since 389 is inert in $\mathbb{Q}(\sqrt{-3})$,*

$$\text{div}(j) = 3 \sum_{i=1}^{130} (Q_i) - (P_\infty) - 389(P_0).$$

Both cusps map to O_E . Indeed, their images are rational torsion points, and

$$\#E(\mathbb{F}_5) = 9, \quad \#E(\mathbb{F}_7) = 13$$

imply $E(\mathbb{Q})_{\text{tors}} = 0$. Pushing this principal divisor forward to E gives

$$3 \sum_{i=1}^{130} \varphi(Q_i) = O_E.$$

The sum is rational, so it is O_E . Consequently, if ι is the nontrivial automorphism of $F/\mathbb{Q}$, then

$$S_H^\iota = -S_H, \quad \text{Tr}_{F/\mathbb{Q}}(S_H) = O_E.$$

For this Weierstrass equation, $-(x, y) = (x, -1 - y)$. Thus the displayed value gives the rational point

$$\left(\frac{83054760289201}{68231958132816}, \frac{591325257523727551823}{5558102574231821778144} \right)$$

on the quadratic twist model

$$389V^2 = 4U^3 + 4U^2 - 8U + 1.$$

8 Some properties of $F_N(x, j)$ and $f_N(x, J)$

We normalize $F_N(x, j)$ by requiring $\gcd\{c_{k,l}\} = 1$ and by choosing the sign so that the leading nonzero coefficient is positive in the lexicographic order with the first variable preceding the second in order to ensure the uniqueness of $F_N(x, j)$. The same restrictions apply to the polynomials $f_N(x, J)$, $G_N(y, j)$, and $g_N(y, J)$. This allows us to state the following proposition. Let Λ_E be the period lattice of E with respect to the associated invariant differential ω_E .

Proposition 8.1. *If $\gamma(0) \in \Lambda_E$, then $F_N(X, Y) = f_N(X, Y)$.*

Proof. Let $f|W_N = \varepsilon f$ where $\varepsilon = \pm 1$. Then we have

$$f\left(\frac{-1}{N\tau}\right) = \varepsilon N\tau^2 f(\tau).$$

Thus

$$\begin{aligned} \gamma\left(\frac{-1}{N\tau}\right) &= \int_{i\infty}^{-\frac{1}{N\tau}} 2\pi i f(z) dz = \int_0^\tau 2\pi i f\left(\frac{-1}{Nu}\right) \frac{1}{Nu^2} du \\ &= \varepsilon \int_{i\infty}^\tau 2\pi i f(u) du - \varepsilon \int_{i\infty}^0 2\pi i f(u) du. \end{aligned}$$

Consequently, we obtain

$$x\left(\frac{-1}{N\tau}\right) = \wp \circ \gamma\left(\frac{-1}{N\tau}\right) - \frac{b_2}{12} = \wp\left(\varepsilon \int_{i\infty}^\tau 2\pi i f(u) du\right) - \frac{b_2}{12} = x(\tau)$$

by $\int_{i\infty}^0 2\pi i f(u) du \in \Lambda_E$. The equality $F_N(X, Y) = f_N(X, Y)$ follows from

$$j\left(\frac{-1}{N\tau}\right) = J(\tau) \quad \text{and} \quad x\left(\frac{-1}{N\tau}\right) = x(\tau).$$

□

Let $C_{x,j}$ be the normalization of the closure of the image of (x, j) in $\mathbb{P}^1 \times \mathbb{P}^1$, and put

$$r = [\mathbb{C}(X_0(N)) : \mathbb{C}(x, j)].$$

The induced map $\rho : X_0(N) \rightarrow C_{x,j}$ has degree r . The tower law gives the exact formulas

$$K = \frac{\mu}{r}, \quad L = \frac{2d}{r}. \tag{9}$$

In particular, $r = 1$ is exactly the case in which the coordinate relation is a birational plane model of $X_0(N)$. For every $R \in C_{x,j}$,

$$\sum_{Q \in \rho^{-1}(R)} e_\rho(Q) = r.$$

There are r distinct points in a general fibre, but special fibres may have fewer points, with the difference accounted for by local multiplicities. A singular point of the plane image can also have several preimages on $C_{x,j}$; these branches must be separated before applying the degree formula.

Lemma 8.2. *Let $N = 4m$ and suppose*

$$\gamma\left(\frac{2}{N}\right) \in \Lambda_E.$$

Then

$$x\left(\frac{\tau}{2m\tau+1}\right) = x(\tau)$$

for all $\tau \in \mathcal{H}$.

Proof. Write $T_{-1/2}(\tau) = \tau - 1/2$, and let $w = \begin{pmatrix} 1 & 0 \\ N/2 & 1 \end{pmatrix}$. Since $4 \mid N$, the newform satisfies $f|U_2 = 0$, so $f(\tau - 1/2) = -f(\tau)$. Moreover, $f|W_N = \epsilon f$ with $\epsilon = \pm 1$, and $W_N T_{-1/2} W_N$ represents w up to a scalar matrix. The weight-two slash action therefore gives

$$f|_2 w = -f.$$

Integrating and using $w\infty = 2/N$, we obtain

$$\gamma(w\tau) + \gamma(\tau) = \gamma(2/N).$$

The hypothesis implies $\gamma(w\tau) \equiv -\gamma(\tau) \pmod{\Lambda_E}$. The evenness and periodicity of $\wp_{\Lambda_E}$ now give $x(w\tau) = x(\tau)$. $\square$

Remark 8.3. *Assume that $4 \mid N$, and put*

$$w = \begin{pmatrix} 1 & 0 \\ N/2 & 1 \end{pmatrix}.$$

Then $w \in \Gamma_0(N/2)$. Moreover, w normalizes $\Gamma_0(N)$. Thus w induces an automorphism of $X_0(N)$. Since

$$w^2 = \begin{pmatrix} 1 & 0 \\ N & 1 \end{pmatrix} \in \Gamma_0(N),$$

this automorphism is an involution. Under $\gamma(2/N) \in \Lambda_E$, we have $x(w\tau) = x(\tau)$. Since x is already invariant under $\Gamma_0(N)$, and since

$$[\Gamma_0(N/2) : \Gamma_0(N)] = 2, \quad w \in \Gamma_0(N/2) \setminus \Gamma_0(N),$$

it follows that x is invariant under $\Gamma_0(N/2)$. Therefore x descends to a meromorphic function on $X_0(N/2)$.

Recall that $E \cong \mathbb{C}/\Lambda_E$ and $\{M_n \mid 1 \leq n \leq \mu\}$ is a complete set of right coset representatives for $\Gamma_0(N)$ in $\text{SL}_2(\mathbb{Z})$.

Proposition 8.4. *If $4 \mid N$ and $\gamma\left(\frac{2}{N}\right) \in \Lambda_E$, then the degrees of $F_N(x, j)$ satisfy*

$$K \leq \frac{\mu}{2} \quad \text{and} \quad L \leq d.$$

Proof. The preceding involution is nontrivial and fixes both x and j . Consequently the index r in (9) is at least two. The formulas $K = \mu/r$ and $L = 2d/r$ prove the two inequalities. $\square$

Example 8.5. *Consider the elliptic curve 40a1 defined by:*

$$y^2 = x^3 - 7x - 6.$$

The genus g of $X_0(40)$ is 3, $\mu = [\text{SL}_2(\mathbb{Z}) : \Gamma_0(40)] = 72$, $\deg(\varphi) = 2$, $K = \deg_x F_{40}(x, j) = 36 = \frac{\mu}{2}$, $L = \deg_j F_{40}(x, j) = \deg(\varphi) = 2$. Thus (9) gives $r = 2$, and the map to the normalization of the (x, j) -relation curve has degree two. The integral points on 40a1 are $(-2, 0)$, $(-1, 0)$, and $(3, 0)$. We observe that the equation $F_{40}(-2, j) = 0$ has no solutions. Similarly, $F_{40}(-1, j) = 0$ and $F_{40}(3, j) = 0$ have no solutions. In fact, $\varphi^{-1}((-2, 0)) = \{[\frac{1}{4}], [\frac{1}{8}]\}$, $\varphi^{-1}((3, 0)) = \{[1], [\frac{1}{2}]\}$, $\varphi^{-1}((-1, 0)) = \{[\frac{1}{5}], [\frac{1}{10}]\}$, $\varphi^{-1}(O_E) = \{[\frac{1}{20}], [\frac{1}{40}]\}$. Take the points $P_1 = (-3, 2\sqrt{3}i)$ and $P_2 = (-3, -2\sqrt{3}i)$. The equation $F_{40}(-3, j) = 1073741824 \cdot (11441195895676649024 + 944485450025040j + 847288609443j^2) = 0$ has two solutions

$$j_1 = \frac{10648(-182511805 - 236108339i\sqrt{2})}{3486784401},$$

$$j_2 = \frac{10648(-182511805 + 236108339i\sqrt{2})}{3486784401}.$$

For the approximate τ -values

$$\begin{aligned}\tau_{11} &\approx -0.272844389235320 + 0.00701737923645263i, \\ \tau_{12} &\approx 0.117863782898542 + 0.00292282078245356i, \\ \tau_{21} &\approx -0.250000000000000 + 0.0813850458467884i, \\ \tau_{22} &\approx 0.134961072356135 + 0.0122394381578248i,\end{aligned}$$

the reported numerical fibre assignments are described by

$$j(\tau_{11}) = j(\tau_{12}) = j_1, \quad j(\tau_{21}) = j(\tau_{22}) = j_2,$$

and

$$\varphi^{-1}(P_1) = \{[\tau_{12}], [\tau_{21}]\}, \quad \varphi^{-1}(P_2) = \{[\tau_{11}], [\tau_{22}]\}.$$

The index $r = 2$ counts preimages of a generic point on the normalization of the (x, j) -relation curve. At a special point, one must count each normalization branch and its local multiplicity. It is not a count of all points with a given j -value, which instead has total degree μ . In the displayed pairs $(-3, j_1), (-3, j_2)$, the j -values avoid $0, 1728, \infty$; the degree-two map is therefore unramified above their normalization branches. The particular numerical representatives still require the image-recognition check of Section 3 for exact fibre labels.

9 A variant of Algorithm 1

Since

$$\mathbb{Q}(X_0(N)) = \mathbb{Q}(j, J), \quad J(\tau) = j(N\tau),$$

the coordinate functions of the modular parametrization, which is defined over $\mathbb{Q}$, admit expressions

$$x = \frac{P_1(j, J)}{Q_1(j, J)}, \quad y = \frac{P_2(j, J)}{Q_2(j, J)}, \quad P_i, Q_i \in \mathbb{Q}[U, V],$$

where $Q_i(j, J) \neq 0$. Such rational expressions appear in Kolyvagin's work [17]. After a common scalar multiplication of each pair, we may take P_i, Q_i to have integer coefficients. The expressions are identities of meromorphic functions; direct pointwise evaluation requires the relevant denominator to be nonzero.

For an algebraic approach, regard

$$F_N(T, j), \quad f_N(T, J)$$

as polynomials in T over the function field $\mathcal{K} = \mathbb{Q}(X_0(N)) = \mathbb{Q}(j, J)$. Here j and J satisfy $\Phi_N(j, J) = 0$; they are not independent indeterminates. Since $x \in \mathcal{K}$, the linear polynomial $T - x$ divides their greatest common divisor. If this greatest common divisor has degree one, a nonzero linear subresultant has the form $a(j, J)T + b(j, J)$, with $a(j, J) \neq 0$, and gives $x = -b(j, J)/a(j, J)$. Clearing denominators then gives $Q_1(j, J)x - P_1(j, J) = 0$. The degree-one condition must be checked: the existence of the factor $T - x$ does not exclude further common factors. If the greatest common divisor has larger degree, the desired linear factor must be selected separately, for example by the q -expansion certificate below. The same qualification applies to the two relations for y .

For computational purposes, we recover the rational expressions directly from q -expansions by solving a linear system. Two checks are required: the proposed identity must hold in the function field, and its denominator must represent a nonzero function. The following lemma supplies finite certificates for both checks.

Lemma 9.1 (Finite certificate for rational-function recovery). *Let $d = \deg \varphi$ and $\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$. Let $P, Q \in \mathbb{Q}[U, V]$ have bidegrees bounded by (K, L) and (R, S) , respectively, and put*

$$\begin{aligned}h &= P(j, J) - xQ(j, J), \\ A_x &= \max\{K + NL, 2 + R + NS\}, \quad B_x = 2d + \mu(\max\{K, R\} + \max\{L, S\}), \\ B_Q &= \mu(R + S).\end{aligned}$$

In the normalization of Section 2, if

$$[q^n]h = 0 \quad (-A_x \leq n \leq B_x),$$

then $h \equiv 0$. Moreover,

$$Q(j, J) \not\equiv 0 \iff [q^n]Q(j(q), J(q)) \neq 0 \text{ for some } -R - NS \leq n \leq B_Q.$$

Consequently, these two finite checks certify $x = P(j, J)/Q(j, J)$.

Proof. The pole divisor of h is bounded by

$$(h)_\infty \leq \max\{K, R\}(j)_\infty + \max\{L, S\}(J)_\infty + (x)_\infty.$$

Since

$$\deg(j) = \deg(J) = \mu, \quad \deg(x) = 2d,$$

we have $\deg(h)_\infty \leq B_x$. The local pole orders of j, J, x at P_∞ are $1, N, 2$, so h has no terms below q^{-A_x} . The asserted coefficient vanishing gives $\text{ord}_{P_\infty} h \geq B_x + 1$. If h were nonzero, its zero divisor would then have greater degree than its pole divisor, a contradiction.

Similarly,

$$(Q(j, J))_\infty \leq R(j)_\infty + S(J)_\infty, \quad \deg(Q(j, J))_\infty \leq B_Q.$$

There are no local terms below q^{-R-NS} . If all coefficients through q^{B_Q} vanish, the same divisor-degree argument proves $Q(j, J) \equiv 0$. Conversely, an identically zero function has every coefficient zero. $\square$

An alternative exact denominator test uses

$$\mathcal{A} = \mathbb{Q}[U, V]/(\Phi_N(U, V)), \quad \text{Frac}(\mathcal{A}) = \mathcal{K}.$$

Namely,

$$Q(j, J) \not\equiv 0 \iff Q \notin (\Phi_N).$$

Because Φ_N is monic in V , this can be checked by taking the remainder of Q on division by Φ_N in $\mathbb{Q}[U][V]$. A nonzero formal polynomial alone does not suffice: a multiple of Φ_N represents the zero function. The finite coefficient test in the lemma avoids computing Φ_N solely for this purpose. If reduced representatives replace the original polynomials, their bidegrees must be used when applying the bounds above.

Algorithm 6 Certified recovery of $x = P_1(j, J)/Q_1(j, J)$

Input: Exact q -expansions of x, j, J available to arbitrary precision, and N, d, μ .

Output: $P_1, Q_1 \in \mathbb{Z}[U, V]$ with $Q_1(j, J) \not\equiv 0$ and $P_1(j, J) = xQ_1(j, J)$.

- 1: Set $m = 0$.
- 2: **loop**
- 3: Set $K = L = R = S = m$. Form A_x, B_x, B_Q as in Lemma 9.1, and set $M = B_x$.
- 4: Compute the Laurent coefficients of $j^k J^l$ and $xj^r J^s$, for the stated bidegree ranges, through exponent M inclusively, with sufficient input precision.
- 5: Compute a basis over $\mathbb{Q}$ of the solutions of

$$\sum_{k=0}^K \sum_{l=0}^L c_{k,l}[q^n](j^k J^l) - \sum_{r=0}^R \sum_{s=0}^S d_{r,s}[q^n](xj^r J^s) = 0, \quad -A_x \leq n \leq M.$$

- 6: **for** each basis vector **do**
 - 7: Form $P_1(U, V) = \sum c_{k,l}U^kV^l$ and $Q_1(U, V) = \sum d_{r,s}U^rV^s$.
 - 8: **if** some coefficient of $Q_1(j(q), J(q))$ in the range $-R - NS \leq n \leq B_Q$ is nonzero **then**
 - 9: Clear rational denominators simultaneously in P_1, Q_1 .
 - 10: **return** (P_1, Q_1) .
 - 11: **end if**
 - 12: **end for**
 - 13: Set $m = m + 1$.
 - 14: **end loop**
-

The lemma proves correctness of every returned pair. Inspecting a basis of the solution space is sufficient, since the map sending a coefficient vector to the function $Q_1(j, J)$ is linear. If this map vanishes on every basis vector, it vanishes on the entire solution space. Finally, $x \in \mathbb{Q}(j, J)$ implies that a representation with a nonzero denominator exists with finite bidegrees. The displayed degree schedule eventually contains such a pair, proving termination. Other exhaustive schedules for (K, L, R, S) may be used.

Remark 9.2 (Construction and input precision). *The number M in Algorithm 6 is the largest exponent tested, so the linear system has $M + A_x + 1$ coefficient equations before dependent rows are removed. The number of formal unknowns is $(K + 1)(L + 1) + (R + 1)(S + 1)$.*

For candidate generation one may use a smaller empirical cutoff, including the original choice

$$M_0 = \max\{(K + 1)(L + 1), (R + 1)(S + 1)\} + 50.$$

Before accepting a candidate, however, both finite checks in the lemma must succeed. If the candidate fails, extend the system at the same degree bounds through B_x and recompute its kernel before enlarging the degree bounds. A nonzero kernel vector alone is not an acceptance criterion. In PARI/GP, if $\mathcal{U}$ and $\mathcal{V}$ are the two coefficient matrices, `matker(concat( $\mathcal{U}$ ,  $-\mathcal{V}$ ))` returns the coefficient kernel directly; `concat` here forms the horizontal block concatenation. Using `matintersect` instead requires recovering the coefficient vectors mapping to a common image.

To account for negative powers, put

$$\hat{x} = q^2 x, \quad \hat{j} = qj, \quad \hat{J} = q^N J.$$

These are power series. To compute every required product through exponent M , it is sufficient to know all three normalized series modulo

$$q^{T_x}, \quad T_x = M + A_x + 1.$$

Equivalently, sufficient absolute input precisions are

$$x(q) \pmod{O(q^{T_x-2})}, \quad j(q) \pmod{O(q^{T_x-1})}, \quad J(q) \pmod{O(q^{T_x-N})}.$$

Here $O(q^t)$ means that coefficients with exponent strictly less than t are known. A product with total local pole order at most A_x then has error $O(q^{M+1})$. These are sufficient bounds; they need not be optimal.

For the y -coordinate the same argument applies, with

$$A_y = \max\{K + NL, 3 + R + NS\}, \quad B_y = 3d + \mu(\max\{K, R\} + \max\{L, S\}).$$

The denominator bound remains $B_Q = \mu(R + S)$. Use $q^3 y$ in place of $q^2 x$ and $T_y = M + A_y + 1$ for the normalized input precision.

Applying Algorithm 6 to elliptic curves E with conductors $N \in \{11, 14, 17, 19\}$, we obtain modular parametrizations expressed as rational functions:

$$x = \frac{P_1(j, J)}{Q_1(j, J)}, y = \frac{P_2(j, J)}{Q_2(j, J)}.$$

Substituting these expressions into the Weierstrass equation of E and clearing denominators produces an integer-coefficient polynomial in j and J that must be divisible by the modular polynomial $\Phi_N(j, J)$. This computational approach, while effective for small conductors, faces significant limitations - as pointed out by Yang in [23], it is difficult to explicitly write down these modular functions in general.

The same recovery method applies to a Yang-pair (X, Y) [23, 7] generating $\mathbb{Q}(X_0(N))$, provided that its pole degrees and local pole bounds are known. For the x -coordinate, the global certificate is

$$B_x^{X,Y} = 2d + \max\{K, R\} \deg(X) + \max\{L, S\} \deg(Y),$$

and the denominator bound is

$$B_Q^{X,Y} = R \deg(X) + S \deg(Y).$$

If $a, b \geq 0$ bound the pole orders of X, Y at P_∞ , the local starting exponent is $-A_x^{X,Y}$, where

$$A_x^{X,Y} = \max\{aK + bL, 2 + aR + bS\}.$$

The normalized input series are $q^2 x, q^a X, q^b Y$; the same precision argument applies. For y , replace $2d$ by $3d$ and the local contribution 2 by 3 . The following example illustrates the resulting rational expressions.

Example 9.3. *Consider the elliptic curve 46a1 defined by:*

$$y^2 + xy = x^3 - x^2 - 10x - 12.$$

The genus for $X_0(46)$ is $g = 5$. Define the modular functions

$$X := \frac{1}{2} \sum_{11} \frac{E_1 E_{14} E_{15} E_{16} E_{17} E_{18}}{E_5 E_6 E_7 E_8 E_9 E_{22}} - \frac{19}{2},$$

$$Y := \sum_{11} \frac{E_{16} E_{21}}{E_2 E_7} - 2X - 19,$$

where

$$E_g(\tau) = q^{\frac{g^2}{92} - \frac{g}{2} + \frac{23}{6}} \prod_{n=1}^{\infty} (1 - q^{46(n-1)+g})(1 - q^{46n-g}), \quad q = e^{2\pi i \tau},$$

and $\sum_{11} \prod E_g^{e_g}$ denotes $\sum_{\gamma \in \Gamma \setminus \Gamma_0} \prod E_g^{e_g} | \gamma$, where Γ is $\langle \Gamma_1(46), -I \rangle$, so that $[\Gamma_0(46) : \Gamma] = 11$.

Define

$$P_1(X, Y) := X^4(8Y - 1341) + X^3(12Y^2 - 2278Y - 25484) +$$

$$X^2(12Y^3 - 1482Y^2 - 30222Y - 114264) +$$

$$X(4Y^4 - 541Y^3 - 16698Y^2 - 76176Y) +$$

$$(-74Y^4 - 4393Y^3 - 42849Y^2),$$

$$Q_1(X, Y) := -387X^4 + X^3(4Y^2 - 275Y - 6785) +$$

$$X^2(-54Y^2 - 3726Y - 28566) +$$

$$X(24Y^3 - 1311Y^2 - 9522Y) + (12Y^4 + 69Y^3 - 4761Y^2).$$

Then $x = P_1(X, Y)/Q_1(X, Y)$. The functions are defined over $\mathbb{Q}$. Here $\Gamma = \langle \Gamma_1(46), -I \rangle$, and the trace is $\sum_{\gamma \in \Gamma \setminus \Gamma_0(46)} F|_0 \gamma$. The modularity and cusp-order formulas for generalized eta products [23] show that the summands are holomorphic away from cusps lying above ∞ on $X_0(46)$. More explicitly, for either exponent vector (e_g) appearing above, the order bound at a cusp with $\delta = \gcd(c, 46)$ is

$$\frac{\delta}{2} \sum_g e_g B_2 \left(\left\{ \frac{ag}{\delta} \right\} \right), \quad B_2(t) = t^2 - t + \frac{1}{6},$$

for the diamond representatives $a = 1, 3, \dots, 21$. These values are zero for $\delta = 1, 2, 23$. At ∞ , the trace expansions begin

$$X = q^{-6} - q^{-5} + 2q^{-4} - q^{-3} + q^{-2} + 3q^{-1} - 8 + O(q),$$

$$Y = q^{-7} - 2q^{-6} + 3q^{-5} - 4q^{-4} + 4q^{-3} - 2q^{-2} - 6q^{-1} + 1 + O(q).$$

Thus $(X)_\infty = 6(P_\infty)$ and $(Y)_\infty = 7(P_\infty)$. Rationality of the trace expansions gives rationality over $\mathbb{Q}$ by the q -expansion principle. The index $[\mathbb{Q}(X_0(46)) : \mathbb{Q}(X, Y)]$ divides both 6 and 7, and hence is one.

Their defining equation is

$$\Psi_{46}(U, V) = U^7 + 8U^6 - 26VU^5 - (22V^2 + 184V)U^4$$

$$- (23V^3 + 115V^2)U^3 - (12V^4 + 138V^3)U^2$$

$$- (5V^5 + 46V^4)U - V^6 - 23V^5.$$

Exact expansion gives $\Psi_{46}(X(q), Y(q)) = O(q^{198})$. Its global pole-degree bound is $7 \cdot 6 + 6 \cdot 7 = 84$, so the identity is certified. Since X, Y generate the function field, the bidegrees $(7, 6)$ also identify this as their irreducible relation.

For the displayed P_1, Q_1 , take $K = L = R = S = 4$. Here $d = 5$, so the sufficient certificate bound is

$$B_x^{X,Y} = 10 + 4 \cdot 6 + 4 \cdot 7 = 62.$$

The verification script `verify46_yang_certificate.gp` constructs the trace expansions directly from the eta products and checks

$$P_1(X(q), Y(q)) - x(q)Q_1(X(q), Y(q)) = O(q^{206}),$$

$$Q_1(X(q), Y(q)) = 4q^{-32} + O(q^{-31}).$$

Thus the denominator is a nonzero function and the displayed rational expression for x is an exact identity. The constant $-19/2$ in the definition of X is needed for these expansions and the stated coefficients of P_1, Q_1 . The example uses rational recovery in a Yang-pair without requiring an involutory factorization of φ . It does not assert an obstruction to other constructions using Yang-pairs.

10 From rational points on E to algebraic points on $Z_0(N)$

Let $\Phi_N(X, Y)$ denote the classical modular polynomial. Define the plane curve

$$Z_0(N) := \{(X, Y) \in \mathbb{C}^2 : \Phi_N(X, Y) = 0\},$$

which serves as the canonical plane model for $X_0(N)$. Let W be the finite set of singular points of $Z_0(N)$. The canonical map

$$\pi : Y_0(N) \rightarrow Z_0(N), \quad \tau \mapsto (j(\tau), J(\tau))$$

induces a bijection between $Y_0(N) \setminus \pi^{-1}(W)$ and $Z_0(N) \setminus W$. Over a point of W , the fiber of the normalization map may contain more than one point.

For an algebraic point $(j_0, J_0) \in Z_0(N)$, we define its degree by

$$\deg(j_0, J_0) = [\mathbb{Q}(j_0, J_0) : \mathbb{Q}].$$

Thus (j_0, J_0) is called a degree d point if

$$[\mathbb{Q}(j_0, J_0) : \mathbb{Q}] = d.$$

If $[\tau]$ is a degree d point of $X_0(N)$, then $(j(\tau), J(\tau))$ is a point of $Z_0(N)$ with the degree at most d [1].

Given a finite point $(j_0, J_0) \in Z_0(N)$, the equations

$$F_N(T, j_0) = 0, \quad f_N(T, J_0) = 0$$

provide candidates for the finite x -coordinates of points of $X_0(N)$ lying above (j_0, J_0) . A nonsingular plane point has a unique preimage under normalization, but these two equations alone need not identify its coordinate uniquely without an additional uniqueness argument. The correct coordinate can be selected by a certified rational expression whose denominator is nonzero at the point, or by recognition on $X_0(N)$. Points mapping to O_E are treated separately.

Proposition 10.1. *Let $P = (\alpha, \beta) \in E(\mathbb{Q})$, and suppose that $\varphi([\tau]) = P$ with $\tau \in \mathcal{H}$. Then*

$$[\mathbb{Q}(j(\tau), J(\tau)) : \mathbb{Q}] \leq d,$$

where $d = \deg \varphi$ and $J(\tau) = j(N\tau)$. In particular,

$$[\mathbb{Q}(j(\tau)) : \mathbb{Q}] \leq d, \quad [\mathbb{Q}(J(\tau)) : \mathbb{Q}] \leq d.$$

Proof. Put $Q = [\tau]$. Since

$$\varphi : X_0(N) \longrightarrow E$$

is a nonconstant morphism of smooth projective curves over $\mathbb{Q}$, it is finite of degree $d = \deg \varphi$. Since $P \in E(\mathbb{Q})$, the scheme-theoretic fiber $\varphi^{-1}(P)$ is a zero-dimensional scheme over $\mathbb{Q}$ of degree d . Hence, for every point Q in this fiber, its residue field $\kappa(Q)$ satisfies

$$[\kappa(Q) : \mathbb{Q}] \leq d.$$

The point $Q = [\tau]$ is non-cuspidal because $\tau \in \mathcal{H}$. Thus the modular functions j and $J = j(N\tau)$ are regular at Q . Since both j and J are rational functions on $X_0(N)$ defined over $\mathbb{Q}$, their values at Q lie in $\kappa(Q)$. Therefore

$$\mathbb{Q}(j(\tau), J(\tau)) \subseteq \kappa(Q).$$

It follows that

$$[\mathbb{Q}(j(\tau), J(\tau)) : \mathbb{Q}] \leq [\kappa(Q) : \mathbb{Q}] \leq d.$$

The two individual degree bounds follow immediately. $\square$

Corollary 10.2. *Assume that $\deg \varphi = 1$ and that*

$$x = \frac{P_1(j, J)}{Q_1(j, J)}, \quad y = \frac{P_2(j, J)}{Q_2(j, J)}$$

with $P_i, Q_i \in \mathbb{Z}[j, J]$. For any point $Q = [\tau]$ outside the cusps and the zero loci of $Q_1(j, J)Q_2(j, J)$, one has

$$\varphi(Q) \in E(\mathbb{Q}) \iff (j(\tau), J(\tau)) \in \mathbb{Q}^2.$$

Proof. If $(j(\tau), J(\tau)) \in \mathbb{Q}^2$, then the displayed rational expressions give

$$x(Q), y(Q) \in \mathbb{Q},$$

because both evaluated denominators are nonzero at Q . Hence $\varphi(Q) \in E(\mathbb{Q})$.

Conversely, suppose that $\varphi(Q) \in E(\mathbb{Q})$. Since $\deg \varphi = 1$, the morphism

$$\varphi : X_0(N) \rightarrow E$$

is an isomorphism over $\mathbb{Q}$. Therefore $Q \in X_0(N)(\mathbb{Q})$. Since j and J are rational functions on $X_0(N)$ defined over $\mathbb{Q}$, it follows that

$$j(\tau), J(\tau) \in \mathbb{Q}.$$

□

Remark 10.3. *The proposition is a direct consequence of the degree of a finite morphism, rather than a new degree theorem for modular curves. Its computational use here is that the coordinate polynomials provide equations for locating the bounded-degree points.*

If $P \in E(\mathbb{Q})$ has infinite order, choose $Q_n \in \varphi^{-1}(nP)$ for each $n \geq 1$. Cusp images are torsion, so all Q_n are non-cuspidal, and

$$[\mathbb{Q}(j(Q_n), J(Q_n)) : \mathbb{Q}] \leq d.$$

The Q_n are distinct because their images nP are distinct. The normalization map to $Z_0(N)$ has finite fibres, so their pairs $(j(Q_n), J(Q_n))$ form an infinite set as well. This supplies a concrete way to record modular data associated with rational points. Possible relations with canonical heights are left for future work; no regulator formula or consequence for higher-rank BSD is claimed here.

A Power-basis construction and root selection for 389a1

The compact certificate of Section 7 is sufficient for verifying the sum. We retain the following representation details to specify consistently the original construction.

A single primitive element. Let $a > 0$ and $b > 0$ be the leading coefficients of f_0 and g_0 , respectively. Put

$$X_i = ax_i, \quad Y_i = by_i, \quad s = \sqrt{-3}, \quad \theta = X_1 - s.$$

The numbers X_i, Y_i, θ are algebraic integers. We use this single generator throughout, with minimal polynomial

$$h(T) = \text{Res}_U(f_1(U), (U - T)^2 + 3), \quad f_1(U) = a^{129}f_0(U/a).$$

In fact, $\mathbb{Q}(\theta) = L$. To prove this, suppose that $\rho \in G$ fixes θ . If $\rho(s) = s$, then $\rho(X_1) = X_1$, and hence $\rho = 1$. If $\rho(s) = -s$, put $\eta = \rho c \in G_0$. Since $c(X_1) = X_1$, we obtain

$$\eta(X_1) = X_1 - 2s.$$

Iterating this identity contradicts the finite order of η . Thus the stabilizer of θ is trivial. It follows that h is monic, irreducible, and of degree 260. Its roots are

$$\{ax_i \pm \sqrt{-3} : 1 \leq i \leq 130\}.$$

For $\sigma_i = \sigma^{i-1}$, choose k with $\sigma_k = \sigma_i^2$. For every $r \geq 0$, the trace identity used in the construction is

$$\text{Tr}_{L/K}(X_i\theta^r + X_i(X_k + s)^r) = \text{Tr}_{L/\mathbb{Q}}(X_i\theta^r) \in \mathbb{Z}.$$

Indeed, $cX_i = \sigma_i^{-1}X_1$, and invariance of the relative trace under σ_i^2 gives

$$\text{Tr}_{L/K}(c(X_i\theta^r)) = \text{Tr}_{L/K}(X_i(X_k + s)^r).$$

The same argument applies to Y_i . This explains the extra symmetry required for the displayed relative trace to be an integer. It gives the power-basis representations

$$X_i = \sum_{r=0}^{259} a_{i,r}\theta^r, \quad Y_i = \sum_{r=0}^{259} b_{i,r}\theta^r, \quad a_{i,r}, b_{i,r} \in \mathbb{Q}.$$

If numerical approximations are used to recover the integer traces, their error bounds must isolate a unique integer. Numerical recognition alone supplies candidate representations. All subsequent reductions in this basis use the same polynomial h .

Analytic root-selection certificate. At $\tau_0 = (-1 + \sqrt{-3})/2$, put $q = -e^{-\pi\sqrt{3}}$. The Hasse bound and multiplicativity give

$$|a_n| \leq d(n)\sqrt{n} \leq 2n, \quad \left| \gamma(\tau_0) - \sum_{n=1}^{12} \frac{a_n}{n} q^n \right| \leq \frac{2|q|^{13}}{1 - |q|}.$$

Outward rational bounds, using Machin's formula for π and Taylor bounds for the exponential, give

$$-\frac{4352144615873901098808832377}{10^{30}} \leq \gamma(\tau_0) \leq -\frac{4352144615873901098808832373}{10^{30}}.$$

On the real branch near O_E with $y_E > 0$, the elliptic logarithm is

$$w(t) = -I(t), \quad I(t) = \int_t^\infty \frac{du}{\sqrt{4u^3 + 4u^2 - 8u + 1}}.$$

For $t \geq 52794$, elementary binomial bounds give

$$\frac{1}{\sqrt{t}} \left(1 - \frac{1}{6t} + \frac{1}{5t^2} - \frac{1}{56t^3} \right) \leq I(t) \leq \frac{1}{\sqrt{t}} \left(1 - \frac{1}{6t} + \frac{1}{5t^2} - \frac{1}{56t^3} + \frac{3}{40t^2} \right).$$

These inequalities give the stated interval for x_1 . The Weierstrass equation and the positive branch then give the stated interval for y_1 .

Remark A.1. *The function `bestappr` can be used to propose a rational candidate from a numerical approximation. Knowing in advance that the desired value is rational does not by itself certify this reconstruction. A proof requires a denominator bound together with a sufficiently small error bound, or an independent exact verification. Here the final value is certified by a principal-divisor identity.*

Declarations

No funding was received for conducting this study.

Competing interests

The author declares that he has no competing interests.

Data and code availability

The article-specific repository [26] contains the original construction scripts and polynomial data. The version referenced here is commit `720e2ddf35f603d09722fa2890b8ce37d6716d66` (17 August 2025). Its data entry points are `F389xj.nb` and the archive *Computational results for Algorithm 1.zip*. These legacy scripts perform construction and numerical exploration; they should be distinguished from the certified procedures specified in this version of the article.

The accompanying verification supplement, version 2026-09-08, contains four standalone PARI/GP scripts, their run transcripts, the detailed 389a1 certificate documentation, and a SHA-256 manifest. Its README specifies the input assumptions and the scope of every check. The new verification runs used PARI/GP 2.15.2. The original construction was reported with PARI/GP 2.17; the archived notebook format does not by itself identify the Mathematica version used. The new supplement is separate from the fixed repository commit above.

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